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(54) **THERAPEUTIC COMPOSITIONS FOR
TREATING PAIN VIA MULTIPLE TARGETS**

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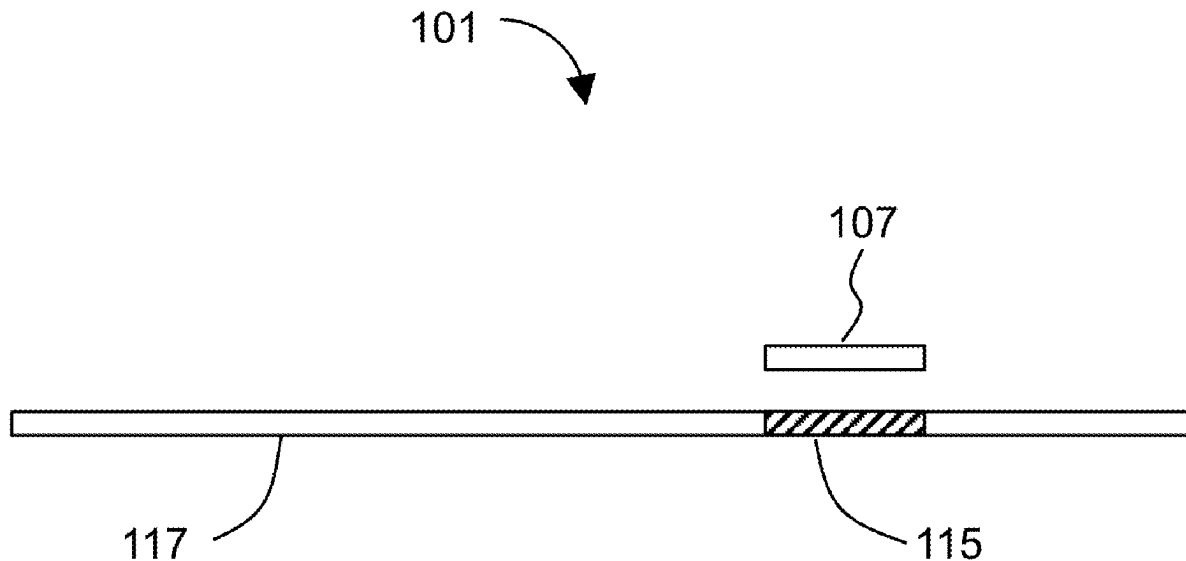
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(57) **ABSTRACT**

The invention provides non-opioid pain therapeutic compositions that include an antisense oligonucleotide (ASO) complementary to an identified target on a NaV channel mRNA. The ASO hybridizes to its target RNA and forms a duplex that recruits RNase H to degrade the RNA, thereby downregulating NaV channel synthesis, which inhibits the neuron's ability to contribute to the perception of pain. The ASO targets one of the specific identified targets, and may be provided as a gapmer that includes a central DNA segment flanked by modified RNA wings. When the composition is delivered to dorsal root ganglion (DRG) neurons in vitro, the DRG neurons exhibit a dose-dependent knock-down of NaV1.7, NaV1.8, or NaV1.9.

Specification includes a Sequence Listing.



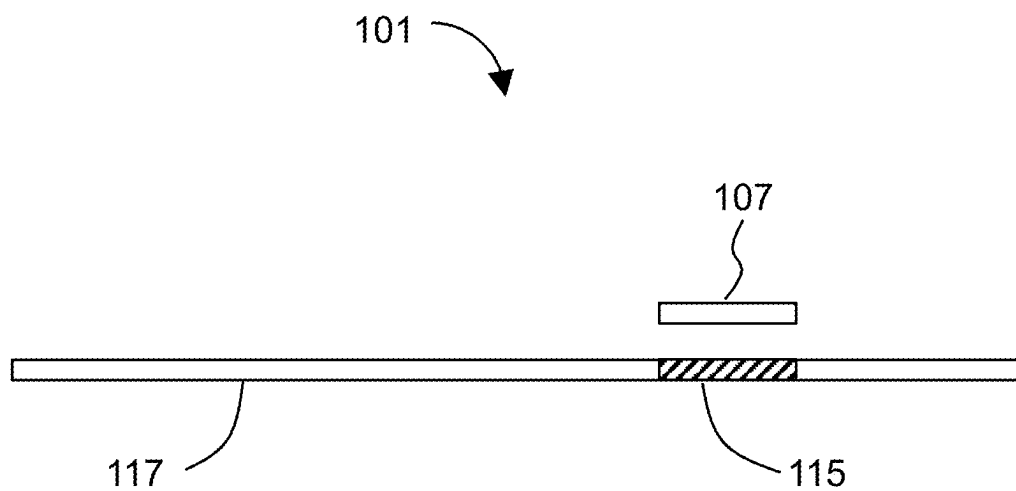


FIG. 1

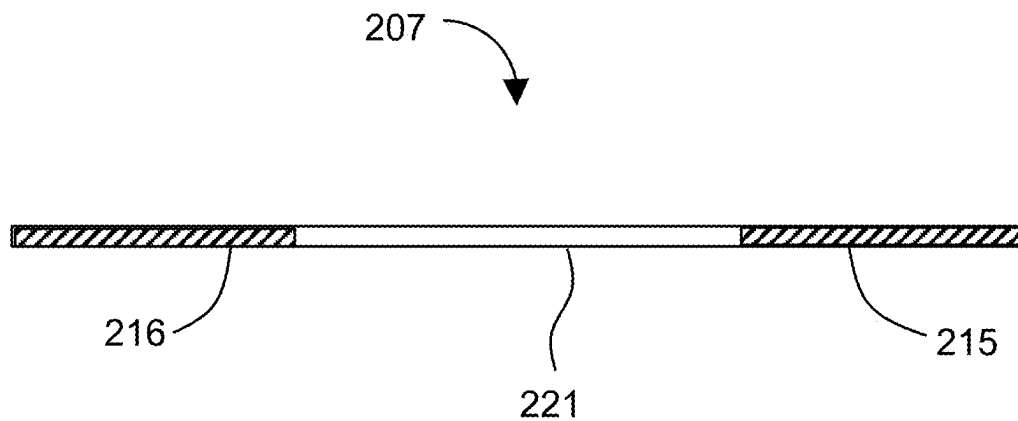


FIG. 2

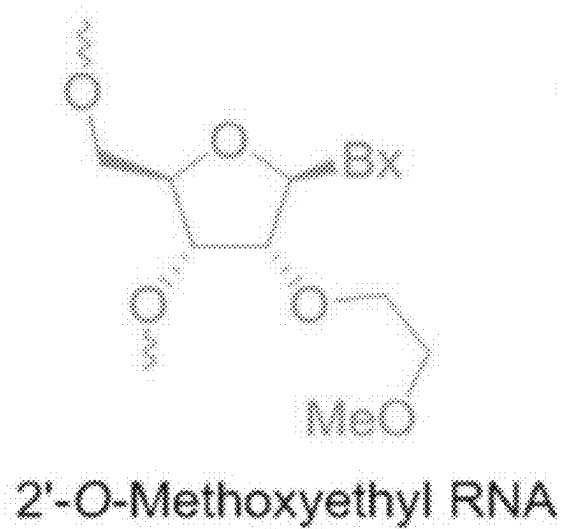


FIG. 3

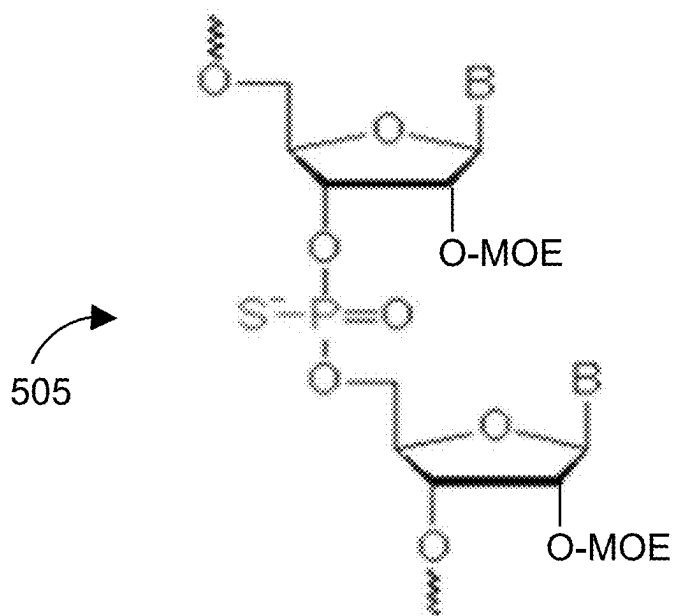


FIG. 4

rNav1.7 knockdown in DRG neurons, qPCR results

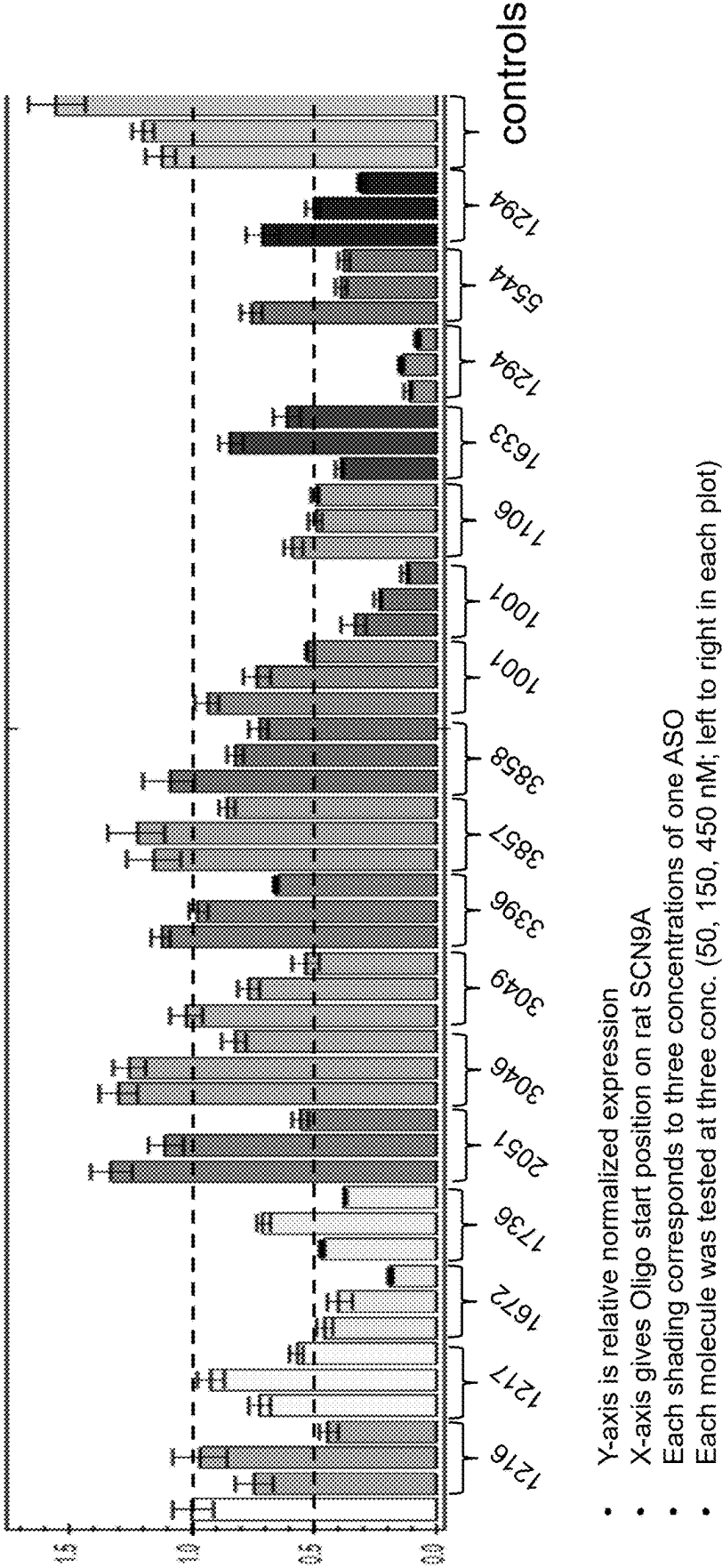
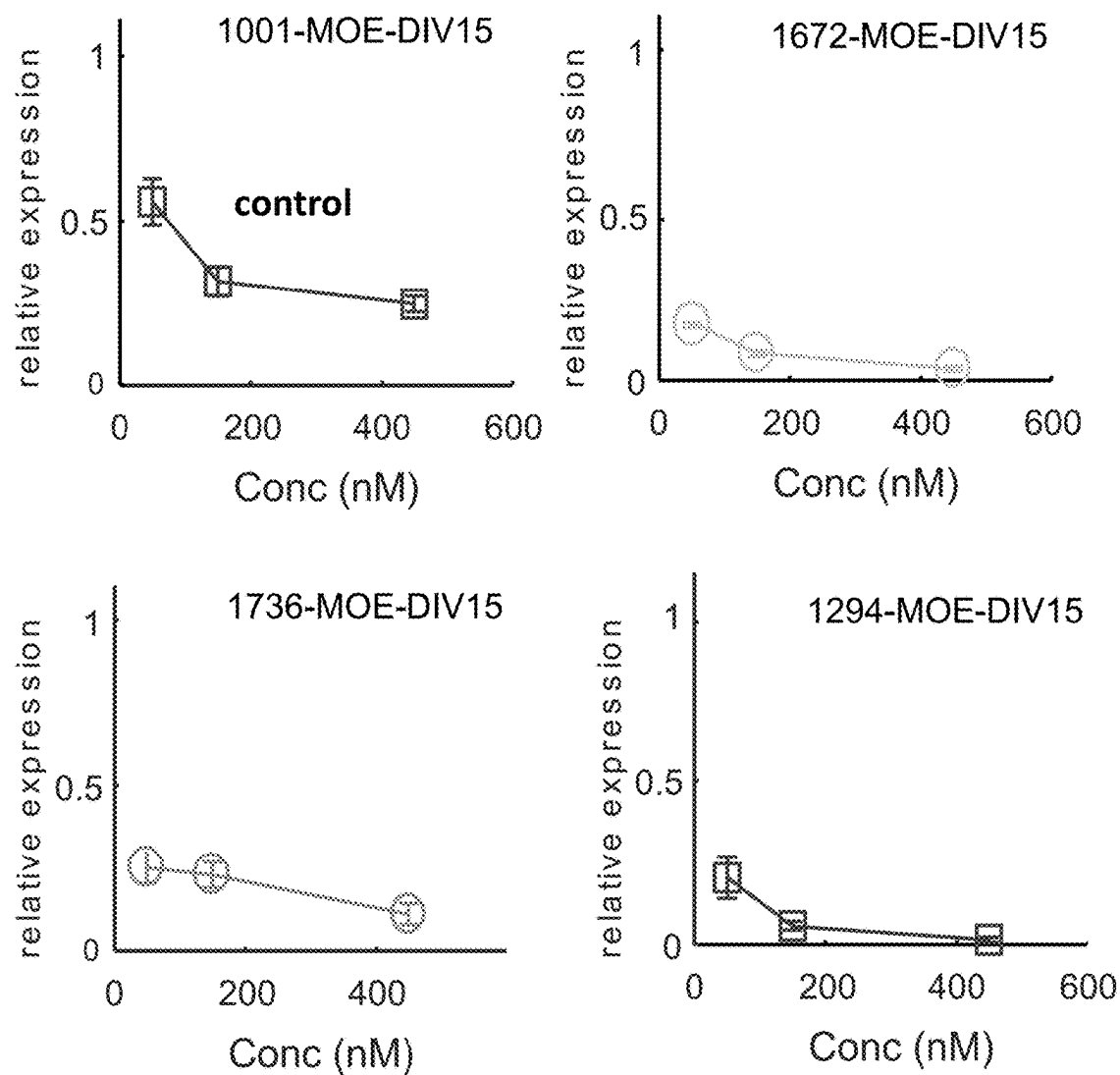


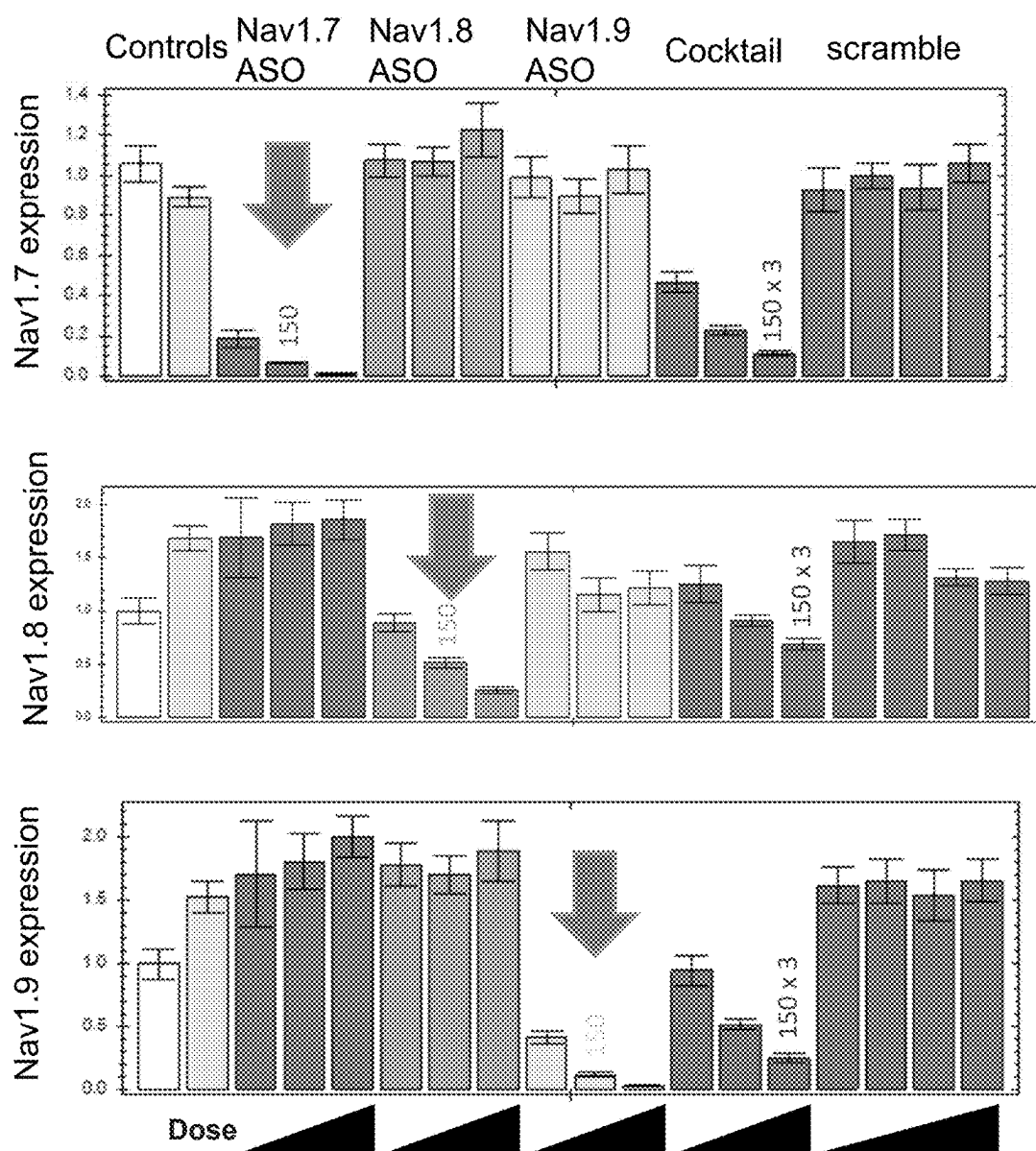
FIG. 5

Nav1.7-targeting ASOs effect on
mRNA expression at DIV14 in rat DRG neurons



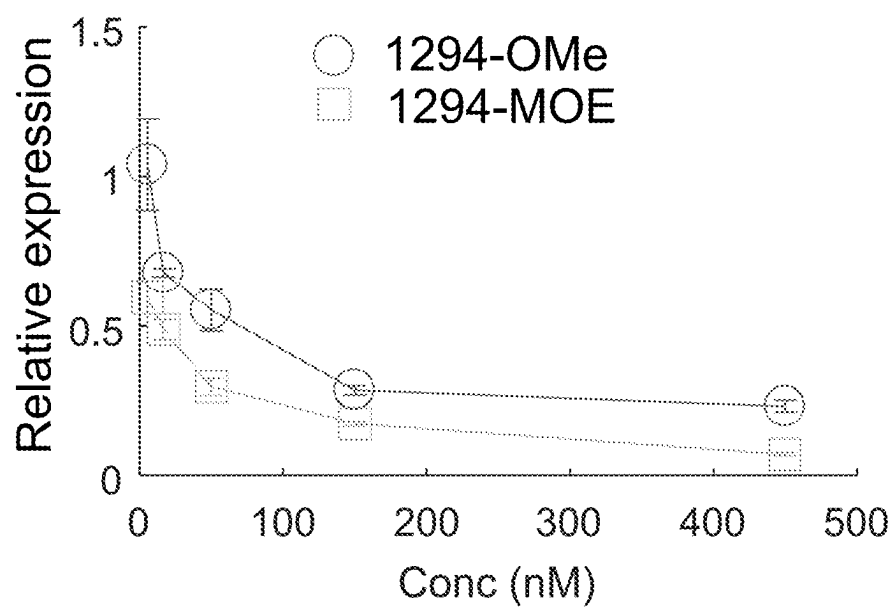
1. 1672-MOE and 1294-MOE show >95% reduction in mRNA in DIV14
2. ASO candidates show better performance than control

FIG. 6



- Nav1.x ASOs have high potency and subtype selectivity
- 450 nM cocktail ASO (150 nM Nav1.7 ASO + 150 nM Nav1.8 ASO + 150 nM Nav1.9 ASO) can knockdown all the three targets

FIG. 7

**FIG. 8**

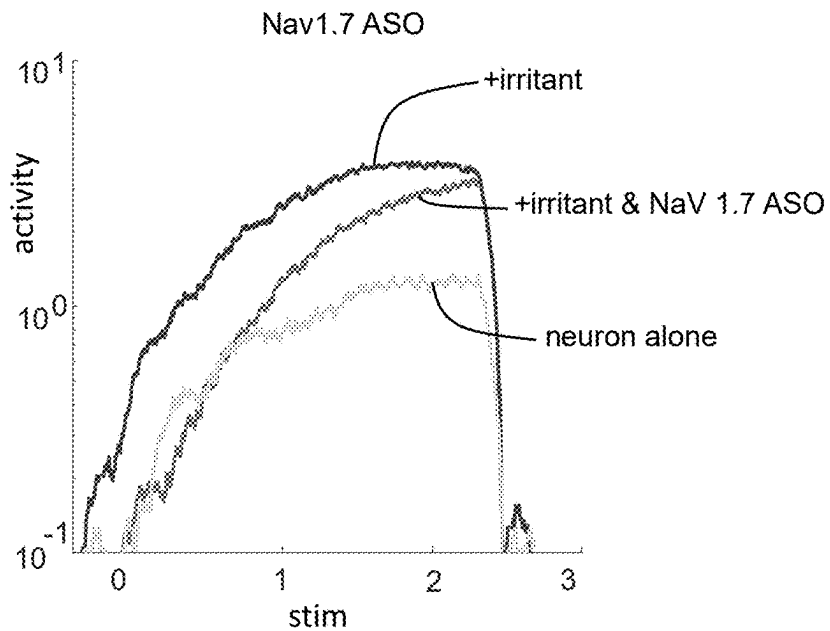


FIG. 9

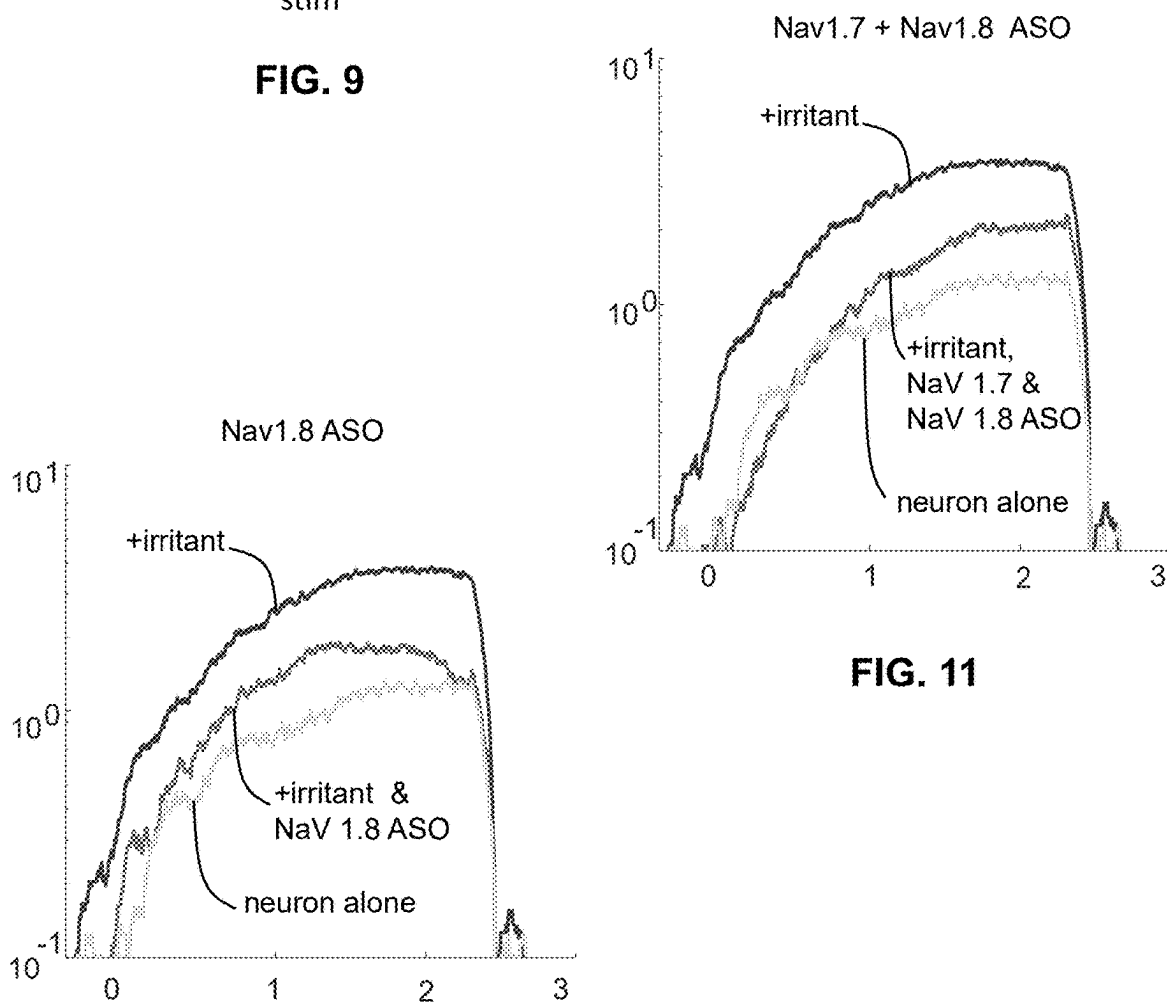


FIG. 10

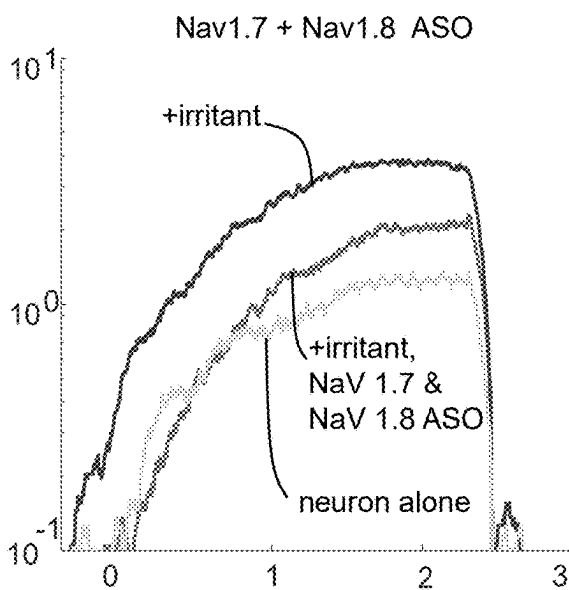


FIG. 11

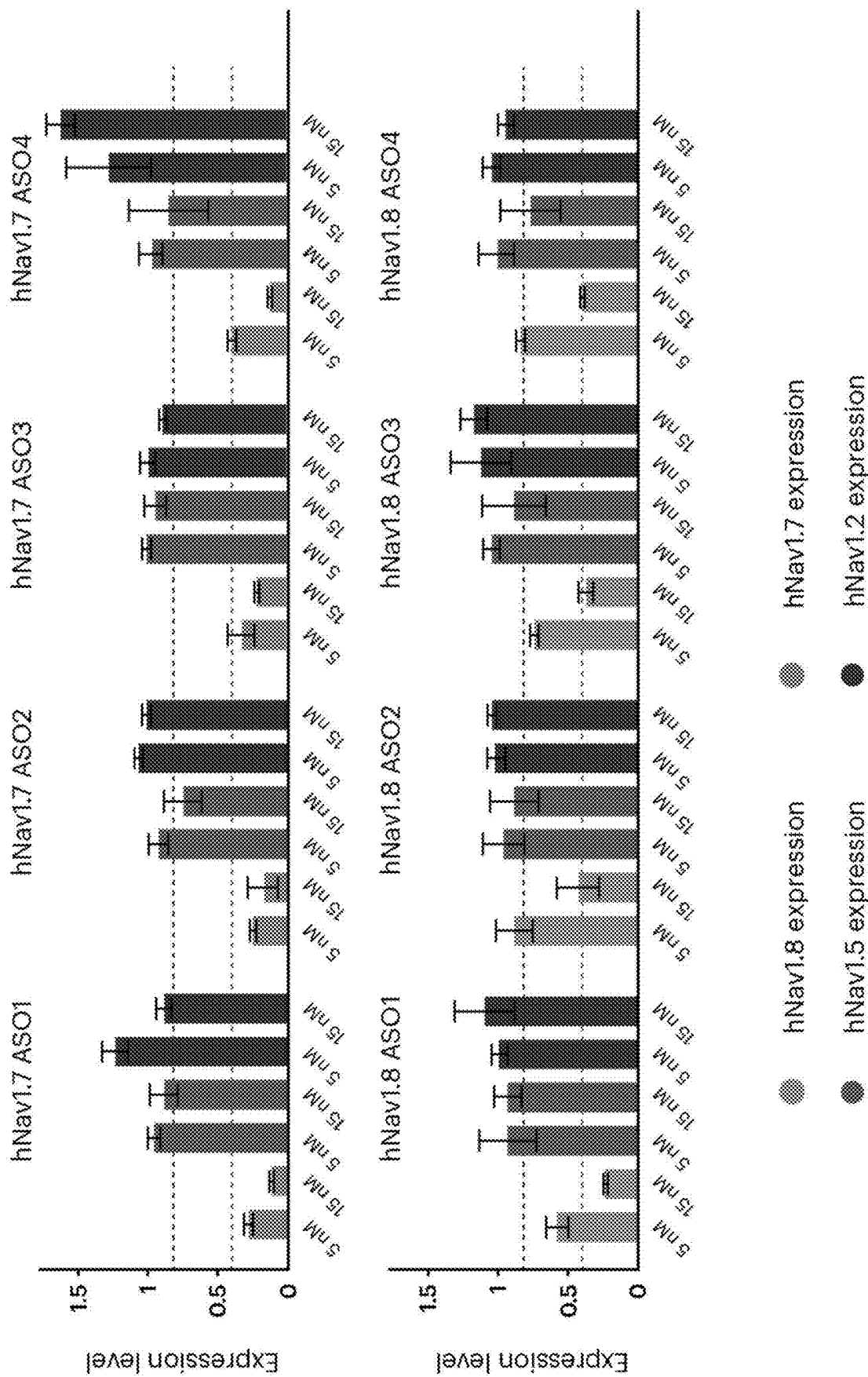


FIG. 12

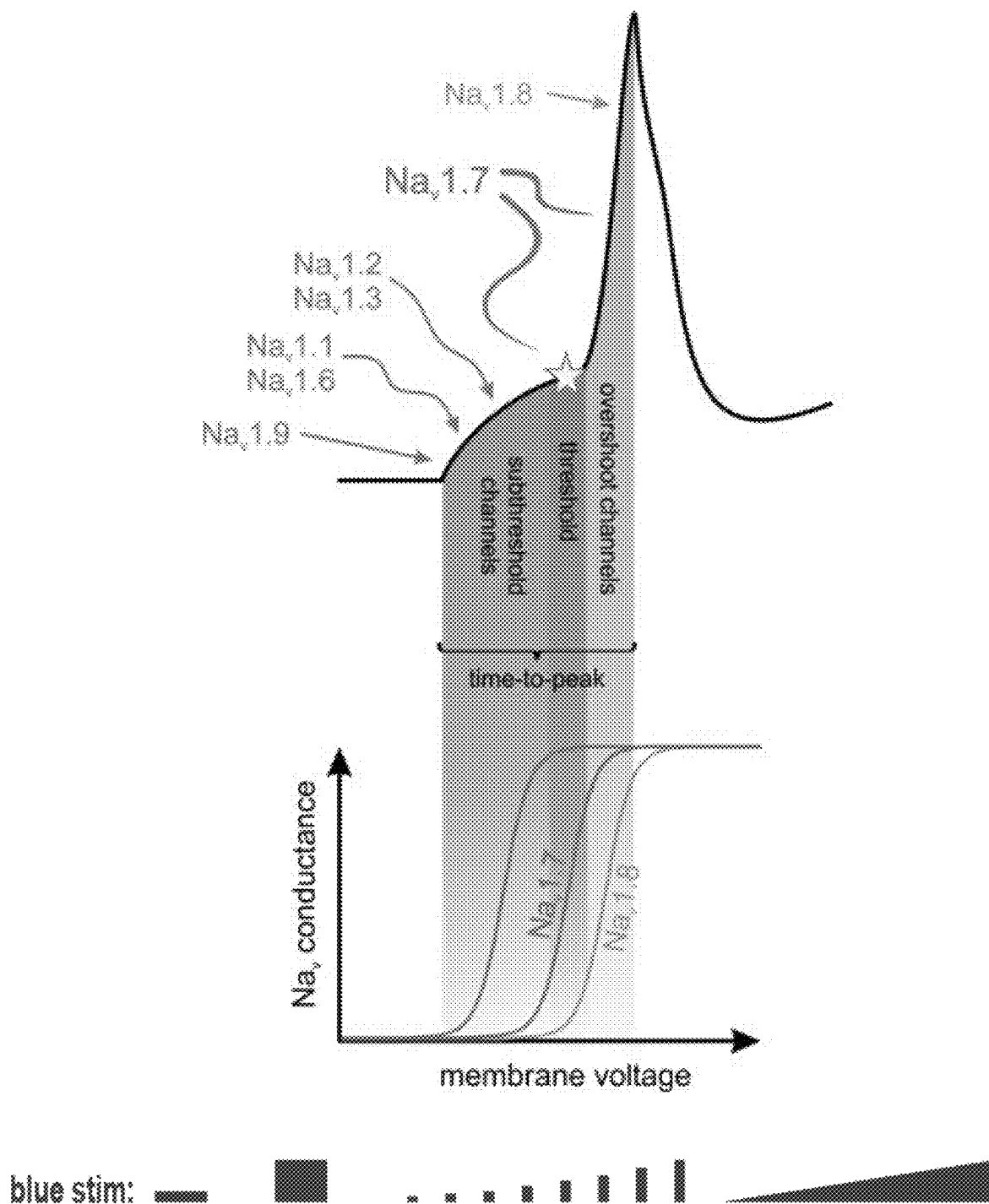


FIG. 13

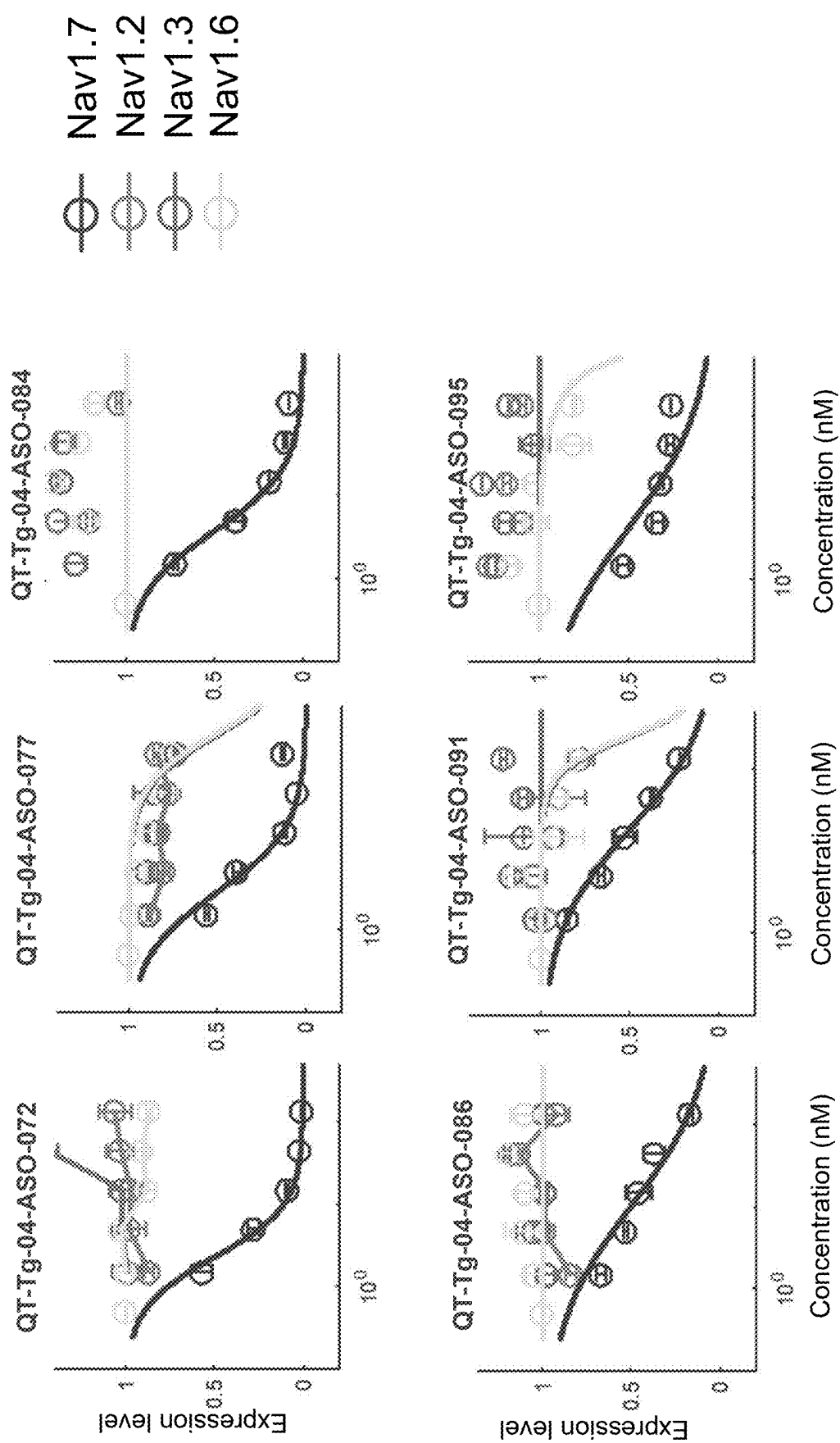


FIG. 14

QuellTx ASO ID	SK-N-AS Nav1.7 IC50 (nM)	SK-N-AS Nav1.2 IC50 (nM)	SK-N-AS Nav1.3 IC50 (nM)	SK-N-AS Nav1.6 IC50 (nM)
QT-Tg-04-ASO-072	2.29	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-077	2.82	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-084	3.76	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-086	10.22	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-091	18.53	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-095	4.32	>200 nM	>200 nM	>200 nM

FIG. 15

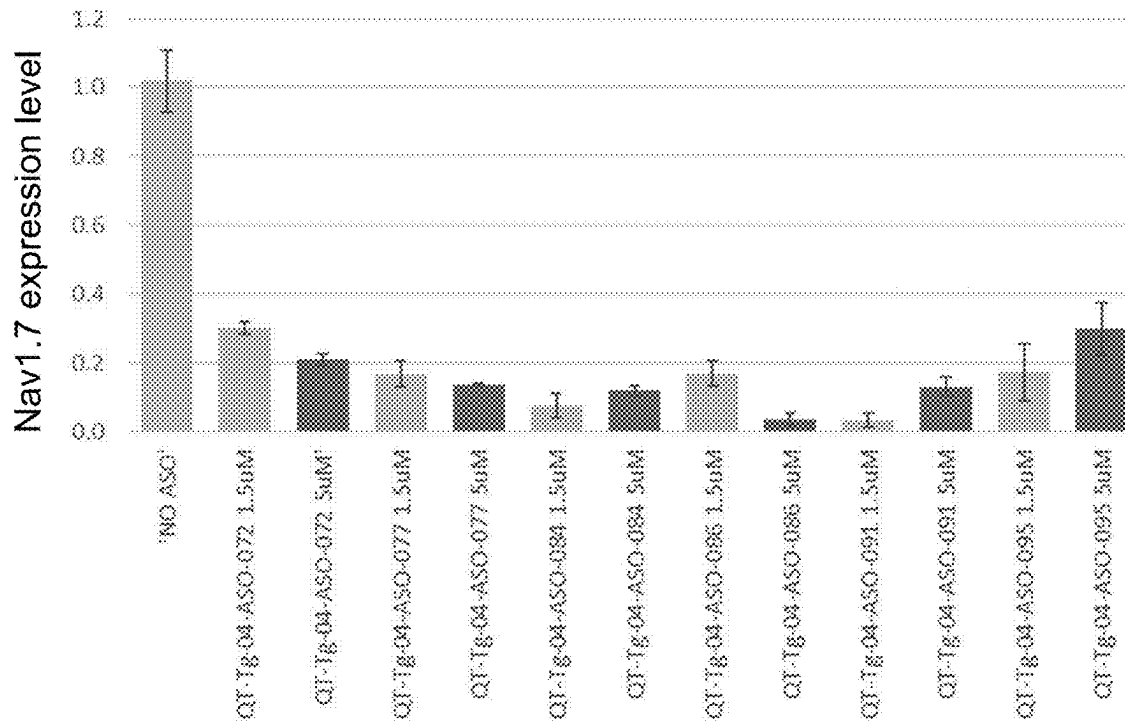


FIG. 16

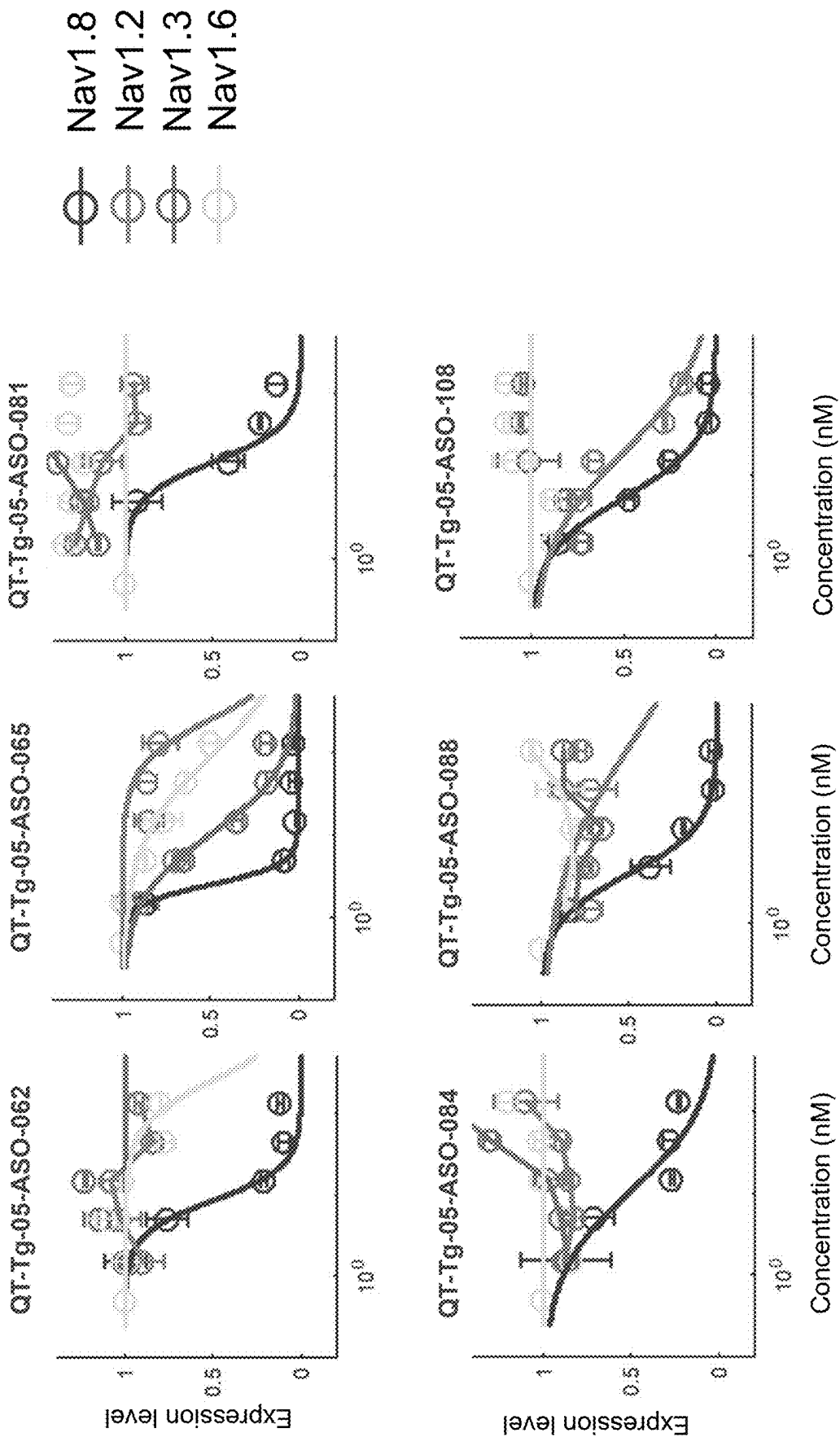


FIG. 17

QuellTx ASO ID	Nav1.8 IC50 (nM)	Nav1.2 IC50 (nM)	Nav1.3 IC50 (nM)	Nav1.6 IC50 (nM)
QT-Tg-04-ASO-062	8.52	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-065	2.51	>200 nM	10.3	92.37
QT-Tg-04-ASO-081	14.39	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-084	10.84	>200 nM	>200 nM	>200 nM
QT-Tg-04-ASO-088	4.12	148.15	>200 nM	>200 nM
QT-Tg-04-ASO-108	5.23	19.01	>200 nM	>200 nM

FIG. 18

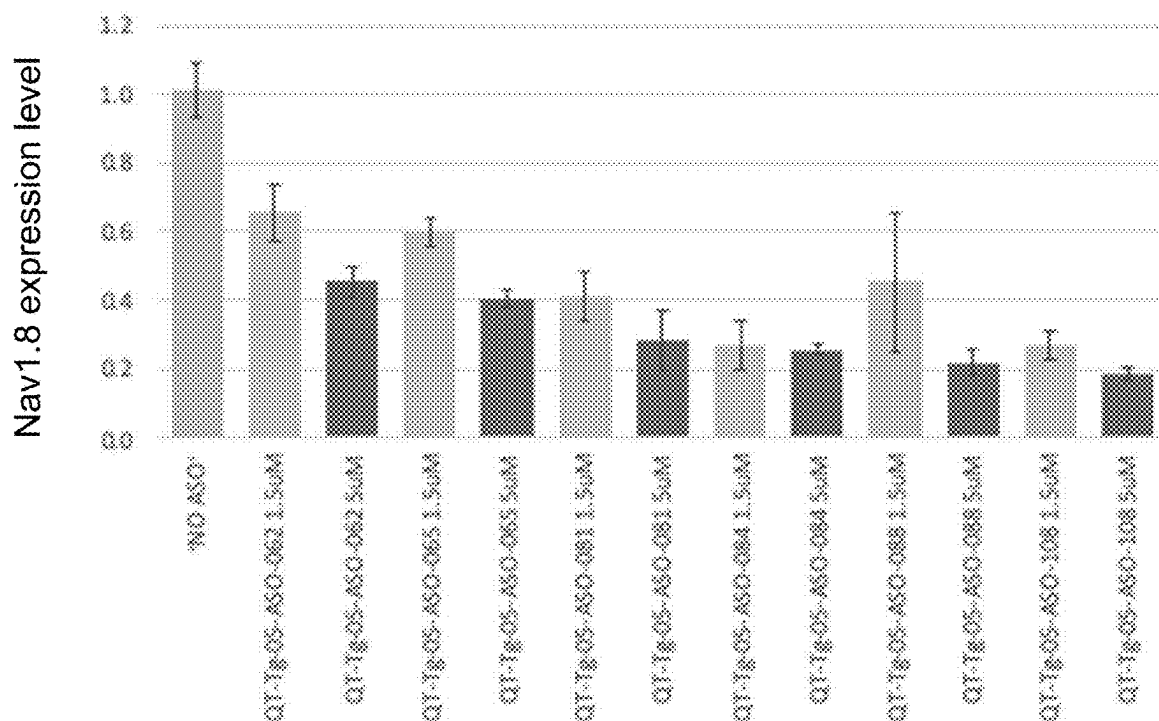


FIG. 19

THERAPEUTIC COMPOSITIONS FOR TREATING PAIN VIA MULTIPLE TARGETS

TECHNICAL FIELD

[0001] The disclosure relates to non-opioid therapeutic compositions for treating pain.

SEQUENCE LISTING

[0002] This application is being filed with a sequence listing in computer readable form in the accompanying XML file named QSTA-29-04US.xml, which was created on May 1, 2025, and is 498 KB in size, the contents of which are hereby incorporated by reference.

BACKGROUND

[0003] In the United States, the Center for Disease Control estimates that as many as 100 million people suffer from chronic pain. One prevalent approach to the treatment of pain involves the use of opioids. Although opioid drugs are highly effective treatments for pain, the abuse of those addictive drugs is understood to be an epidemic problem. Nevertheless, there are many common medical conditions that are painful to experience and live with.

[0004] One common medical condition that often induces severe chronic pain is osteoarthritis. Osteoarthritis occurs when the cartilage in joints breaks down or wears away, eventually leading to exposed bone on the joint surfaces that rub together and can fragment or splinter. Many people who suffer from osteoarthritis are familiar with the enduring pain that this condition brings. Another common medical condition that may cause severe chronic pain is cancer. The American Cancer Society attributes cancer pain to the cancer itself, not merely to an inflammatory response to the cancer. Research indicates that cancer cells themselves drive hypersensitivity of sensory neurons. Thus, not only can a cancer manifest as a tumor, or spread throughout the body, but the cancer itself can be a direct cause of severe pain.

[0005] There are various approaches to treating pain, but each is associated with specific limitations. Over-the-counter non-steroidal anti-inflammatory drugs (NSAIDs) may lack the necessary efficacy to treat certain forms of pain. In addition, some people become non-responsive to NSAIDs or cannot tolerate adverse effects on the digestive system and kidney function. Opioids are understood to be effective for treating pain, but come with a steep human and societal cost and may lose efficacy over time due to development of tolerance. Opioids are addictive narcotics and are well-understood to be implicated in abuse, diversion, and even fraud and criminal activity. Further serious drawbacks of NSAID and opioid treatment include a high death toll.

SUMMARY

[0006] The invention provides therapeutic compositions useful to treat pain that do not require or involve opioids. The compositions include short nucleic acids, or oligonucleotides, which prevent the synthesis of proteins involved in the perception of pain. Specifically, certain neurons operate as “pain-sensing” nerves, or nociceptors. Those pain-sensing neurons have proteins that function as voltage-gated sodium channels. When stimulation of the nerve endings exceeds a threshold voltage (V), the nociceptor neurons conduct sodium ions (Na⁺) across the cell membrane, which can cause the neuron to depolarize in a regenerative fashion

leading to “firing” of propagating electrical signals that underlie the sensation of pain. Compositions of the invention include oligonucleotides that bind to the messenger RNA (mRNA), or precursor mRNA (pre-mRNA) used in making those sodium channel proteins that enable pain sensation. The invention includes the identification of numerous specific validated targets within those RNAs. The oligonucleotides prevent those proteins from being made, which decreases the sensitivity or activity of those pain-sensing neurons. Because activity of the pain sensing neurons is decreased, the patient experiences far less pain. The compositions do not need to include any opioids or other narcotics, and thus are not habit-forming. The compositions may be used in combination with opioids to reduce effective doses of opioids. The compositions thus provide long-term pain relief by downregulating sodium channels in pain-sensing neurons.

[0007] There are nine families of voltage-gated sodium channels in humans, named NaV1.1 through NaV1.9. Of those nine proteins, NaV1.7, NaV1.8, and NaV1.9 are expressed in nociceptor dorsal root ganglion (DRG) neurons and contribute to the perception of pain.

[0008] Oligonucleotides of the disclosure are designed to bind to certain targets in the RNAs used in synthesis of the NaV1.7, NaV1.8, and NaV1.9 proteins. Binding of the oligonucleotides prevents protein synthesis and downregulates expression of the corresponding NaV channel. Specifically, the oligonucleotides have a sequence that is substantially or entirely complementary to one of the identified targets on a NaV channel pre-mRNA or mRNA. That is, the oligonucleotide is antisense to the identified target. When the antisense oligonucleotide (ASO) hybridizes to its target RNA, they form a double-stranded ASO: RNA duplex that recruits an enzyme (RNase H) that degrades a portion of the double-stranded duplex. Degrading the ASO:RNA duplex depletes the neuron of NaV channel mRNA, which decreases the amount of NaV channel synthesized by the cell. Downregulating NaV channel expression interferes with the ability of the neuron to contribute to the sensation of pain.

[0009] Thus, when a composition that includes oligonucleotides that are antisense to the identified targets in NaV1.7, NaV1.8, or NaV1.9 pre-mRNA or mRNA is administered to a patient, that patient will have a diminished experience of pain. Accordingly, compositions of the disclosure are useful to treat pain in patients without requiring the use of opioids and may also minimize or lead to lower use of opioids.

[0010] In certain aspects, the disclosure provides a composition for treating pain. The composition includes an oligonucleotide that hybridizes to a pre-mRNA or mRNA encoding a sodium channel protein along a segment of that RNA that is at least about 75% complementary to one of SEQ ID NOs: 1-164 and 166-400 to thereby prevent translation of the RNA into the sodium channel protein. In preferred aspects, the pre-mRNA or mRNA encoding a sodium channel protein along a segment of that RNA is at least about 75% complementary to one of SEQ ID NOs: 166-400. In even more preferred aspects, the pre-mRNA or mRNA encoding a sodium channel protein along a segment of that RNA is at least about 75% complementary to one of SEQ ID NOs 11, 27, 82, 85, 145, 157, and 166-177.

[0011] The oligonucleotide may hybridize to, and knock-down expression of, one or more of NaV1.7, NaV1.8, and

NaV1.9 pre-mRNA or mRNA. Preferably, a sequence of bases in the oligonucleotide has at least 80% identity to one of SEQ ID NOs: 1-164 and 166-400. For example, the sequence of bases in the oligonucleotide may be at least 90% or 95% identical to one of SEQ ID NOs: 1-101, 142-164, and 201-400, and the oligonucleotide may hybridize to, and induce RNase H cleavage of, either NaV1.7 or NaV1.8 pre-mRNA or mRNA. The composition may include a plurality of therapeutic oligonucleotides each having a base sequence at least 80, 90, 95, or 100% identical to one of SEQ ID NOs: 1-164 and 166-400.

[0012] Therapeutic oligonucleotides of the disclosure may have a gapmer structure that includes a central DNA segment flanked by modified RNA wings. Such a therapeutic oligonucleotide may include two wings flanking a central region of DNA bases (e.g., about 8 to 10 DNA bases). Preferably at least one end of the oligonucleotide comprises modified RNA bases, e.g., any number or any combination of 2'-O-methoxyethyl RNA ("2'-MOE") and/or 2'-O-methyl RNA ("2' O-Me"). In addition, compositions of the invention may be designed to target an exon-exon junction in order to differentially target cytoplasmic versus nuclear mRNA. Thus, ASOs of the invention can be designed to interact with RNA prior to or after splicing, adding specificity and versatility to the compositions.

[0013] The therapeutic oligonucleotide may be provided in a solution or carrier formulated for intrathecal injection, preferably about 3 to 4 times per year. The oligonucleotide may be of any suitable length, e.g., at least about 13 bases, preferably between about 15 and 25 bases. The oligonucleotide may have phosphorothioate bonds in the backbone. In preferred embodiments, the oligonucleotide has a base sequence that has been screened and determined to not meet a threshold match for any long, non-coding RNA or other off-target sequences or transcripts in humans. The oligonucleotide may have a base sequence with 0 mismatches to a homologous segment in a non-human primate genome and no more than about 5 mismatches in a homologous segment in a rodent genome.

[0014] When the composition is delivered to dorsal root ganglion (DRG) neurons in vitro, the DRG neurons exhibit a dose-dependent knockdown of NaV1.7, NaV1.8, or NaV1.9. The oligonucleotide may be a gapmer having a base sequence with at least a 90% match to one of

[0015] SEQ ID NOs: 1-164 and 166-400, with bases linked by phosphorothioate linkages. In preferred aspects, the oligonucleotide is a gapmer having a base sequence with a 90% match to one of SEQ ID NOs: 166-400. In even more preferred aspects, the oligonucleotide is a gapmer having a base sequence with a 90% match to one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177. The linkages may be all phosphorothioate or a mixture of phosphorothioate and phosphodiester bonds. The oligonucleotide may further have a central 10 DNA bases flanked by a 5' wing and a 3' wing, the 5' wing and the 3' wing each comprising five consecutive 2' modified RNA bases. Preferably, the oligonucleotide has a base sequence matching one of SEQ ID NO: 1-164 and 166-400, with bases linked by phosphorothioate linkages, and a structure having central DNA bases flanked by a 5' wing and a 3' wing. The number of RNA bases in the wings and DNA bases in the central segment may be 5-10-5 or 4-12-4, or a similar suitable pattern. The 5' wing and the 3' wing may each include several 2'-MOE RNA bases. In more preferred aspects, such an oligonucleotide has a base

sequence matching one of SEQ ID NOs: 166-400 and even more preferably, one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177. For example, the oligonucleotide may have 5 consecutive 2'-MOE RNA bases in each wing with a central 10 DNA bases (a "5-10-5" structure), with phosphorothioate linkages throughout the central DNA segment and a mixture of phosphorothioate and phosphodiester bonds in the wings.

[0016] In combination embodiments, the invention provides compositions that include a plurality of copies of a plurality of distinct therapeutic gapmers, each according to the descriptions above, in a suitable formulation or carrier.

[0017] Preferably, an oligonucleotide of the disclosure exhibits at least 25% better NaV knockdown than a control gapmer (e.g., in an assay using DRG neurons in vitro, wherein the control gapmer consists of GCCAUAATCCGGGTTUCUGC (SEQ ID NO: 165) linked only by phosphorothioate linkages and further comprising a central 10 DNA bases flanked by a 5' wing of five consecutive 2'-MOE RNA bases and 3' wing of five consecutive 2'-MOE RNA bases).

[0018] Aspects of the disclosure provide a use of an antisense oligonucleotide (ASO) for the manufacture of a medicament for treating pain in a patient. In the use, the ASO has at least about 75% identity with one of SEQ ID NOs: 1-164 and 166-400, and more preferably at least 90% identity, e.g., 95% or 100% identity. In preferred aspects, the ASO has at least about 75% identity with one of SEQ ID NOs: 166-400, and more preferably at least 90% identity, e.g., 95% or 100% identity. In even more preferred embodiments, the ASO has at least about 75% identity with one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177.

[0019] Preferred embodiments use an ASO that is between about 15 and 25 bases in length, preferably between about 18 and 22, or between about 19 and 21 (inclusive). In general, reference to "an ASO" includes numerous copies of substantially identical molecules. Accordingly, "an ASO" may be any number, e.g., hundreds of thousands, or millions, of copies of the indicated ASO. In preferred embodiments, the ASO is 20 bases in length and has the sequence of one of SEQ ID NOs: 1-164 and 166-400 and is used in the manufacture of a medicament for the treatment of pain. The ASO may be provided in any suitable format such as, for example, lyophilized in a tube or in solution in a tube, such as a microcentrifuge tube or a test tube. Preferred embodiments of the use target NaV1.7 and/or NaV1.8. One or more (e.g., two, three, four, or five, or more) ASOs may be used in manufacture of the medicament. The one or more ASOs may hybridize to a target in a NaV1.7 or a NaV1.8 pre-mRNA or mRNA. In certain embodiments of the use, a sequence of bases in the ASO is at least 90% identical to one of SEQ ID NOs: 1-101, 142-164, and 166-400. In embodiments, an ASO may have a gapmer structure with a central DNA segment flanked by RNA wings, e.g., a central region of 10 DNA bases with modified RNA bases on both sides of the central region. Each modified RNA base may be 2'-MOE. Preferably a backbone of the ASO has a plurality of phosphorothioate bonds, e.g. most or all of the sugar linkages may be phosphorothioate in the use embodiments. The corresponding medicament may be formulated for intrathecal delivery. Accordingly, the ASO may initially be in a form suitable for mixing into a formulation suitable for introduction by injection or a pump. For example, the ASO (thousands or millions or more of copies of one ASO) may

be lyophilized in a tube or in solution at a known molality or concentration. The ASO may be dissolved or diluted into a pharmaceutically acceptable composition in which a carrier, such as a solvent and/or excipient, includes the ASO and may be loaded in an IV bag, syringe, or pump. The medicament may be made using more than one ASO, e.g., any combination of 2, 3, 4, or 5, or more. Bases in compositions of the invention may be modified or wobble bases may be used in order to increase the breadth and effectiveness of compositions of the invention. In one example, ASOs for use in the invention may contain methylated bases (e.g., 5-methylcytosine, 5-methyluracil (thymine) and others).

[0020] Compositions of the invention may be formulated to accommodate serial dosing. For example, formulations may provide dosages to be administered at two or more separate times and, optionally, with two or more different ASOs, in order to take advantage of optimal therapeutic windows and to avoid potential competition between ASOs. In addition, compositions of the invention, whether administered serially or not, may interact with more than one target, depending on the composition of the ASOs involved. For example, ASOs may comprise targeted mismatches that allow interaction with multiple targets (both within and across mRNA and pre-mRNA species), thus allowing the associated treatment to impact more than one channel.

BRIEF DESCRIPTION OF THE DRAWINGS

- [0021]** FIG. 1 shows a composition for treating pain.
- [0022]** FIG. 2 shows an oligonucleotide with a gapmer structure.
- [0023]** FIG. 3 shows a 2'-O-Methoxyethyl (MOE) modified ribose sugar.
- [0024]** FIG. 4 shows a phosphorothioate bond in a segment of DNA.
- [0025]** FIG. 5 gives results of a qPCR assay for knock-down of Nav1.7 by 17 different ASOs.
- [0026]** FIG. 6 shows effects of Nav1.7-targeting ASOs compared to a control.
- [0027]** FIG. 7 shows the target selectivity of ASOs of the disclosure.
- [0028]** FIG. 8 compares the effects of ribose sugar modification on ASOs.
- [0029]** FIG. 9 shows effects of an anti-Nav1.7 ASO.
- [0030]** FIG. 10 shows effects of an anti-Nav1.8 ASO.
- [0031]** FIG. 11 shows neural activity levels under treatment with a combination composition.
- [0032]** FIG. 14 is results of Human Nav1.7 ASOs test in SK-N-AS cells
- [0033]** FIG. 15 gives IC50 results.
- [0034]** FIG. 16 shows effects on expression.
- [0035]** FIG. 17 is results of Human Nav1.8 ASOs in SK-N-AS cells.
- [0036]** FIG. 18 shows IC50 for ASOs.
- [0037]** FIG. 19 shows expression.

DETAILED DESCRIPTION

[0038] FIG. 1 shows a composition **101** for treating pain. The composition **101** includes an oligonucleotide **107** that hybridizes to a target segment **115** in an mRNA **117** or a pre-mRNA. The mRNA **117** encodes a sodium channel protein. The segment **115** of the mRNA **117** that includes the target is at least about 75% complementary to one of SEQ

ID NOs: 1-164 and 166-400. Hybridization of the oligonucleotide **107** to the segment **115** of the mRNA **117** prevents translation of the mRNA into the sodium channel protein. The oligonucleotide **107** may hybridize to, and knockdown expression of, one or more of Nav1.7, Nav1.8, and Nav1.9 pre-mRNA or mRNA. Preferably, a sequence of bases in the oligonucleotide has at least 80% identity to one of SEQ ID NOs: 1-164 and 166-400, and more preferably at least 90% identity. In certain preferred aspects, the sequence of the bases of the oligonucleotide has at least 80% identity to one of SEQ ID NOs: 166-400, and more preferably at least 90% identity. In more preferred aspects, the sequence of the bases of the oligonucleotide has at least 80% identity to one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177, and more preferably at least 90% identity.

[0039] In certain embodiments, a sequence of bases in the oligonucleotide is at least 90% identical to one of SEQ ID NOs: 1-101, 142-164, and 166-400, wherein the oligonucleotide can hybridize to, and induce RNase H cleavage of, either Nav1.7 mRNA or Nav1.8 mRNA.

[0040] The oligonucleotide **107** hybridizes to the segment **115** in the mRNA **117** because the oligonucleotide **107** is substantially or entirely antisense to the target segment **115** of the mRNA **117**. In that sense, the composition includes an antisense oligonucleotide (ASO). Compositions **101** include ASOs that bind to target RNA with base pair complementarity and exert various effects, based on the ASO chemical structure and design. Various mechanisms, commonly employed in preclinical models of neurological disease and human clinical trial development, may be employed. Those mechanisms include: RNA target degradation via recruitment of the RNase H enzyme; alternative splicing modification to include or exclude exons, and miRNA inhibition to inhibit miRNA binding to its target.

[0041] Preferred embodiments of the disclosure include ASOs that hybridize to voltage gated sodium channel (Nav channel) pre-mRNA or mRNA and recruit the RNase H enzyme. The RNase H enzyme cleaves the Nav channel RNA, which downregulates expression of the Nav channel protein. Thus, oligonucleotide **107** of the disclosure address Nav channels as targets for pain therapy. The disclosure builds on the insights that clinical and preclinical data support the use of small molecule Nav blockers for pain therapy. For example, IV lidocaine and lidocaine patches have exhibited pain relief effects, which suggest that blocking synthesis of lidocaine's target may provide similar effects. In fact, numerous agents used for neuropathic pain such as tricyclics and selective serotonin reuptake inhibitors have multiple mechanisms of action, but share ability to block Nav channels. The sodium channels Nav1.7, Nav1.8, and Nav1.9 have been implicated in pain. Those proteins provide a genetic link to pain phenotypes. For example, Nav1.7 loss-of-function has been linked to congenital insensitivity to pain (replicated in mouse models). Additionally, Nav1.7 and Nav1.8 gain-of-function have both been linked to excessive pain disorders. Moreover, Nav1.9 gain-of function is linked to neuropathy with indifference to pain. The genetic insights provide rationales for selectively targeting Nav1.7 and Nav1.8 and Nav1.9. Compositions that include anti-Nav ASOs may be administered to a subject to treat or diminish pain. It may be found that anti-Nav ASOs offer benefits over other approaches such as lidocaine because anti-Nav ASOs may be state independent and subtype selective.

[0042] Thus, the disclosure provides a use of an antisense oligonucleotide (ASO) for the manufacture of a medicament for treating pain in a patient. In the use, the ASO has at least about 75% identity with one of SEQ ID NOs: 1-141, and more preferably at least 90% identity, e.g., 95% or greater identity. In more preferred embodiments, the ASO has at least about 75% identity with one of SEQ ID NOs: 166-400, and more preferably at least 90% identity, e.g., 95% or greater identity. In even more preferred embodiments, the ASO has at least about 75% identity with one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177, and more preferably at least 90% identity, e.g., 95% or greater identity.

[0043] Preferred embodiments use an ASO that is between about 15 and 25 bases in length, preferably between about 18 and 22 (inclusive). In general, reference to “an ASO” includes numerous copies of substantially identical molecules. Accordingly, “an ASO” may be more than hundreds of thousands or millions of copies of the defined ASO. In preferred embodiments, the ASO is 20 bases in length and has the sequence of one of SEQ ID NOs: 1-164 and 166-400 and is used in the manufacture of a medicament for the treatment of pain. In more preferred embodiments, the ASO is 20 bases in length and has a sequence of one of SEQ ID NOs: 166-400, and in even more preferred embodiments, one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177.

[0044] The ASO may be provided in any suitable format such as, for example, lyophilized in a tube or in solution in a tube, such as a microcentrifuge tube or a test tube. Preferred embodiments of the use target NaV1.7 and/or NaV1.8. One or more (e.g., two, three, four, or five, or more) ASOs may be used in manufacture of the medicament. The one or more ASOs may hybridize to a target in a NaV1.7 or a NaV1.8 mRNA. In certain embodiments of the use, a sequence of bases in the ASO is at least 90% identical to one of SEQ ID NOs: 1-164 and 166-400. In preferred embodiments of the use, a sequence of bases in the ASO is at least 90% identical to one of SEQ ID NOs: 166-400, and in even more preferred embodiments, one of SEQ ID NOs: 11, 27, 82, 85, 145, 157, and 166-177.

[0045] In embodiments of the use, an ASO may have a gapmer structure with a central DNA segment flanked by RNA wings, e.g., a central region of 10 DNA bases with 5 modified RNA bases on both sides of the central region. Each modified RNA base may be 2'-MOE RNA, 2'-O-Me RNA, or other suitable sugar. Preferably a backbone of the ASO has a plurality of phosphorothioate bonds, either exclusively or also including phosphodiester linkages, e.g., most or all of the sugar linkages may be phosphorothioate in the use embodiments. The medicament may be formulated for intrathecal (IT) delivery. Accordingly, the ASO may initially be in a form suitable for mixing into a formulation suitable for introduction into an intrathecal pump. For example, the ASO (thousands or millions or more of copies of one ASO) may be lyophilized in a tube or in solution at a known molality of concentration. The ASO may be dissolved or diluted into a pharmaceutically acceptable composition in which a carrier, such as a solvent or excipient, includes the ASO and may be loaded in an IV bag, syringe, or intrathecal pump. The medicament may be made using more than one ASO, e.g., any combination of 2, 3, 4, or 5, or more.

[0046] Any ASO(s) described in the use embodiment may be included in a composition of the disclosure. Preferred embodiments of compositions of the disclosure include one

or a plurality of therapeutic oligonucleotides each having a base sequence at least 80% identical to one of SEQ ID NOs: 1-164 and 166-400, wherein each of the therapeutic oligonucleotides has a gapmer structure that comprises a central DNA segment flanked by modified RNA wings, wherein the plurality of therapeutic oligonucleotides is provided in a solution or carrier formulated for intrathecal injection. In more preferred embodiments, the sequence is at least 80% identical to one of SEQ ID NOs: 166-400.

[0047] FIG. 2 shows an oligonucleotide **207** with a gapmer structure. The oligonucleotide **207** includes two wings (first wing **215** and second wing **216**) flanking a central region **221** of about 10 DNA bases. In preferred embodiments, the wings **215**, **216** are all or predominantly RNA bases whereas the central region **221** is all or predominantly DNA bases. Preferably, the wings are all RNA bases (modified or unmodified) and the central region is all DNA bases. In some embodiments, each wing consists of 5 RNA bases, all or most of which are modified RNA bases, e.g., in which each modified RNA base is selected from the group consisting of 2'-O-methoxyethyl RNA and 2'-O-methyl RNA. A modified RNA base may include a substitution on a 2' hydroxyl group of a ribose sugar.

[0048] FIG. 3 shows a 2'-O-Methoxyethyl (“2'-MOE”) modified sugar that may be included in an RNA base.

[0049] The oligonucleotide **207** preferably includes at least about 15 bases, and may include between about 15 about 25 bases. In some embodiments, the oligonucleotide **207** has a backbone comprising a plurality of phosphorothioate bonds.

[0050] FIG. 4 shows a phosphorothioate bond **505** within the backbone of a segment of DNA, such as the central region **221** of the oligonucleotide **207**. The oligonucleotide **207** may include one or any number of the phosphorothioate bonds **505**. For example, every backbone linkage within the oligonucleotide **207** may be phosphorothioate, or most, or about half may be.

[0051] The composition **101** may be formulated for delivery. Accordingly, the oligonucleotide **107** may initially be in a form suitable for mixing into a formulation suitable for introduction into a syringe, bag, or injection pump. For example, the oligonucleotide **107** (thousands or millions or more of copies of one oligonucleotide **107**) may be lyophilized in a tube or in solution at a known molality of concentration. The oligonucleotide **107** may be dissolved or diluted into a pharmaceutically acceptable composition in which a carrier, such as a solvent or excipient, includes the oligonucleotide **107** and may be loaded in an IV bag, syringe, or intrathecal pump. As described, the composition **101** includes at least one oligonucleotide **107** with a sequence that is defined by comparison to one of SEQ ID NOs: 1-164 and 166-400. More preferred compositions include at least one of oligonucleotide **107** with a sequence that is defined by comparison to one of SEQ ID NOs: 166-400. Thus, compositions of the disclosure are defined and illustrated by the identified targets.

[0052] Specifically, the oligonucleotide **107** hybridizes to an mRNA encoding a sodium channel protein along a segment of the mRNA that is at least about 75% complementary to one of SEQ ID NOs: 1-164 and 166-400 to thereby prevent translation of the mRNA into the sodium channel protein. This is accomplished where the oligonucleotide has at least about 75% identity to one of SEQ ID Nos: 1-164 and 166-400, preferably at least about 90% or 95%

identity. In certain embodiments, the oligonucleotide has the sequence of one of SEQ ID Nos: 1-164 and 166-400, although one of skill in the art will understand that oligonucleotides with 90 or preferably 95% identity to a complementary target will still tend to hybridize in a sequence-specific manner to the target. Forming a double stranded structure is energetically favorable enough through Watson-Crick base pairing and base stacking that the double stranded structure can tolerate approximately about 1 mismatched base pair every ten or so. Accordingly, under moderately stringent physiological conditions in a DRG neuron, 95% identity should be effective, especially where an oligonucleotide has a gapmer structure with at least a few modified RNA bases or phosphorothioate backbone linkages to protect the oligonucleotide from enzymatic degradation.

[0053] In fact, a feature and benefit of compositions of the disclosure is that the targets (SEQ ID Nos: 1-164 and 166-400) have been screened to rule out sequences for which the complement is present in molecules other than sodium channel transcripts. For example, the sequences have been screened against databases of RNA transcripts including long, non-coding RNA (lncRNA), and initial sequences that matched non-target sequences were excluded. Thus, ASOs with sequences of SEQ ID Nos. 1-164 and 166-400 when administered to a patient should have a minimized chance of hybridizing to non-target sequences. Accordingly, in preferred embodiments, the oligonucleotide **107** has a base sequence that has been screened and determined to not meet a threshold match for any off-target coding or long, non-coding RNA in humans. A composition or use that meets the criteria stated above should not bind to off-target material such as lncRNA in vivo, as the included sequences have been screened against a database of lncRNA. Sequences of the disclosure have been screened for target specificity. Preferably, the oligonucleotide **107** has a base sequence with 0 mismatches to a homologous segment in a human or non-human primate genome and no more than about 5 mismatches in a homologous segment in a rodent genome.

[0054] When the composition is delivered to a dorsal root ganglion (DRG) neuron in vitro, the DRG exhibits a dose-dependent knockdown of NaV1.7, NaV1.8, or NaV1.9.

[0055] FIG. 5 gives results of a qPCR assay for knockdown of NaV1.7 in rats by 17 different ASOs (at 3 concentrations each) designed according to embodiments of the disclosure (20 bases, 10 base DNA central region flanked by RNA wings with 2'-O modified RNA and phosphorothioate linkages through ASO). Along the bottom of the figure, the "start location" is given, i.e., position within the rat SCN9A mRNA. The right most 3 bars show the expression levels when only controls are administered. All 17 ASOs decreased NaV1.7 expression, relative to controls, for at least the highest concentration. The strongest knockdown was exhibited for the ASO specific to a 20 base target beginning at position 1294. The graph gives results for 17 ASOs, 3 concentrations, applied at DIV8, using gymnotic delivery, with expression levels normalized to GAPDH. The graph shows that a composition **101** of the disclosure exhibits dose-dependent knockdown of NaV1.7.

[0056] Because nucleic acid hybridization has some tolerance for mis-matches, it may be found that an oligonucleotide **107** with a base sequence that is at least a 90% match to one of SEQ ID Nos: 1-164 and 166-400, with bases linked only by phosphorothioate linkages, and in which the

oligonucleotide **107** has a central segment of DNA bases flanked by a 5' wing and a 3' wing (e.g., a 5-10-5 structure in which the 5' wing and the 3' wing each comprise five consecutive 2' modified RNA bases flanking 10 DNA bases, or a 4-12-4 structure, or similar) exhibits dose-dependent knockdown according to the pattern shown in the chart. In some embodiments, the oligonucleotide **107** specifically has a base sequence matching one of SEQ ID Nos: 1-164 and 166-400 (more preferably one of SEQ ID Nos: 166-400), with bases linked by phosphorothioate linkages (optionally with some phosphodiester linkages in the wings), in which the oligonucleotide **107** has a central 10 DNA bases flanked by a 5' wing and a 3' wing, and in which the 5' wing and the 3' wing each include five consecutive 2' MOE RNA bases.

[0057] FIG. 6 shows NaV1.7-targeting ASOs effect on mRNA expression at DIV14 in rat DRG neurons compared to a control. It is noted that a composition **101** of the disclosure may include a plurality of copies of a plurality of distinct therapeutic gapmers of one of the preceding claims in a carrier formulated for intrathecal administration. Preferably, any one or more of the oligonucleotides **107** exhibits at least 25% better NaV knockdown than a control gapmer in an assay using DRGs in vitro, where the candidate oligonucleotides and the controls are linked mostly or only by phosphorothioate linkages and include a central segment of DNA bases flanked by a 5' wing of 2'-MOE RNA bases and 3' wing of 2'-MOE RNA bases.

[0058] Because these compositions are effective at knocking down expression of sodium channels, the compositions of the disclosure may be used to treat patient populations that experience severe intractable pain. Methods of the disclosure include administering to a patient in need thereof any composition of the disclosure to thereby treat or alleviate cancer pain (e.g., from metastatic bone cancer) or neuropathic pain (e.g., small fiber neuropathy associated with gain-of-function NaV mutations), or other populations. Methods of the disclosure may be used to target any NaV channel as a primary target and may additionally include oligos for a secondary target. For example, the primary target may be NaV1.7 (with oligos having substantial identity to one or more of SEQ ID Nos: 1-53, 166-171, 178-191, and 391-400) and secondary target may be NaV1.8 and/or NaV1.9 (with oligos having substantial identity to one or more of SEQ ID Nos: 54-101, 172-177, and 192-200 and/or SEQ ID Nos: 102-141, respectively).

[0059] FIG. 7 shows the selectivity of rat NaV-targeted ASOs tested by qPCR in rat DRG. The top panel shows the effect on mRNA levels of delivery an ASO specific for NaV1.7. The middle panel shows the effect of delivering an ASO specific for NaV1.8; the bottom panel is for NaV1.9. In all panels, the fourth triad of bars gives the results for delivering a 450 nM cocktail ASO (150 nM NaV1.7 ASO+ 150 nM NaV1.8 ASO+150 nM NaV1.9 ASO). In each panel, the non-target NaV channel was not knocked down by the ASO (Nav1.7 ASO did not knock down expression of NaV1.8 or NaV 1.9, etc., showing target selectivity). The cocktail of 3 oligonucleotides **107** is shown to knock down all the three targets.

[0060] FIG. 8 compares the effects of ribose sugar modification within oligonucleotides **107** of the disclosure. An oligonucleotide **107** with a 20 base sequence matching the 20 bases beginning at position 1294 of the human SCN9A gene (SEQ ID NO: 6) was tested in a form with 2'-O-methyl- (OMe) and in a form with 2'-O-Methoxyethyl- (MOE)

ribose sugar modifications. Similar tests were performed against the control oligo SEQ ID NO: 165. It was found that the oligonucleotide of the disclosure outperformed the control and also that, as shown, the MOE modification outperforms (shows a greater inhibition of target relative expression) the OMe modification. Thus it may be preferable for an oligonucleotide of the disclosure to include one or more 2'-O-methoxyethyl (MOE) modifications on RNA ribose sugars. For example, the ASOs may have 5' and 3' wings of about five RNA bases each, and in those wings most or all ribose sugars may be 2' MOE.

[0061] Nav1.7 and Nav1.8 ASO combination effects were tested in rat DRG neurons in vitro. The in vitro neurons included optogenetic constructs that provided neural activation under optical stimulus (e.g., a modified algal channelrhodopsin that causes the neuron to fire in response to light) and optical reporters of neural activity (modified archaerhodopsins that emit light in proportion to neuronal membrane voltage and yield signals of neuronal activity). The in vitro neurons were assayed in a fluorescence microscopy instrument, and were treated with a pain mediator composition (e.g., a simulated “cancer pain soup”) that serves as an irritant causing DRG neurons to fire in a manner similar to the experience of pain in vivo. Any suitable optogenetic constructs, optogenetic microscope, or pain mediator compositions may be used. For example, suitable optogenetic constructs include those described in U.S. Pat. No. 9,594,075, incorporated by reference. Suitable optogenetic microscopes include those described in U.S. Pat. No. 10,288,863, incorporated by reference. Suitable pain mediator compositions include those described in WO 2018/165577, incorporated by reference.

[0062] The in vitro DRG assays involve measuring light from an optogenetic neural sample alone, under increasing optical stimulation. This gives a baseline reading of neural excitability. Then, the neural sample is stimulated with an irritant, here, a pain mediator composition comprising a mixture of cytokines, proteases, pH, necrosis factors, or other factors that may be found in vivo at the site of a painful tumor. Light is measured from the neural sample under treatment with that irritant. Finally, the neural sample is treated with a composition of the disclosure. It is hypothesized that, where the irritant moves measured excitability away from the measured baseline, the oligonucleotides **107** will tend to restore measured excitability towards the baseline. It is further expected that Nav1.7 and Nav1.8 exhibit their effects under different (albeit overlapping) conditions of neural activity. The results show that anti-Nav1.7 and anti-Nav1.8 ASOs in combination mitigate neural response to painful irritants a greater amount and over a greater range of input stimulus levels relative to individual Nav 1.7 or Nav 1.8 ASOs.

[0063] FIG. 9 shows a measured level of neural activity in response to an increasing ramp of blue light stimulation at a baseline, under treatment with the irritant, and under treatment with the irritant and an anti-Nav1.7 ASO. As expected, the Nav1.7 ASO decreases the neural excitability exhibited in response to the irritant.

[0064] FIG. 10 shows a measured level of neural activity in response to an increasing ramp of blue light stimulation at a baseline, under treatment with the irritant, and under treatment with the irritant and an anti-Nav1.8 ASO. As expected, the Nav1.8 ASO decreases the neural excitability

exhibited in response to the irritant, albeit in a different region of the input stimulus power.

[0065] FIG. 11 shows a measured level of neural activity in response to an increasing ramp of blue light stimulation at a baseline, under treatment with the irritant, and under treatment with the irritant and a composition comprising an anti-Nav1.7 ASO and an anti-Nav1.8 ASO. The net pain reducing effect of the combination of oligos is better than either oligo alone. The level of activity the DRG neurons is significantly lower across the full range of input stimulation. It may be found that a living subject experiencing a variety of painful conditions would benefit by a greater range and potency of analgesic effects by being administered a composition comprising different oligos that target different Nav channels.

[0066] Methods and compositions of the disclosure may beneficially be used for delivery of therapeutic oligonucleotides **107** described herein to dorsal root ganglion (DRG) neurons in vivo in a patient suffering pain. Any suitable delivery approach may be used including, for example, systemic delivery (e.g., by injection) or local delivery (e.g., by subcutaneous injections or implantation of a slow-release device). In some embodiments, a composition **101** of the disclosure is delivered by intrathecal injection. Methods of the disclosure may involve delivering a composition of the disclosure by intrathecal injection about every few months, e.g., about 3 or 4 times per year.

[0067] Intrathecal injection refers to a route of administration for drugs via an injection into the spinal canal, or into the subarachnoid space so that it reaches the cerebrospinal fluid (CSF) and is useful in spinal anesthesia, chemotherapy, or pain management applications. Intrathecal delivery avoids the composition from being stopped by the blood brain barrier. Preferably, the composition **101** is compounded in such a formulation so as not to contain any preservative or other potentially harmful inactive ingredients that are sometimes found in standard injectable drug preparations. The composition **101** may be provided in an intrathecal pump, or it may be provided in a formulation suitable for reconstitution in the clinic for delivery via an intrathecal pump. For example, the composition **101** may be in solution (e.g., in saline) at a defined concentration. A technician can mix the concentration with a suitable diluent at the clinic for delivery. In other embodiments, the ASOs are lyophilized or otherwise preserved in a dry, solid form, to be resuspended and appropriately diluted in the clinic. Intrathecal delivery reduces concerns about PK, metabolism, peripheral on/off target effects. Intrathecal administration approaches have been validated for clinical use with other drugs such as Ziconotide and ASOs such as nusinersen. Such methods may be used to deliver compositions **101** of the disclosure directly to the nervous system.

[0068] An oligonucleotide of the disclosure, such as a gapmer, ASO, or therapeutic oligonucleotide **107** in a composition **101** may have a sequence defined with reference to one of the sequences set forth in Table 1. For example, an oligonucleotide of the disclosure may have a sequence that is at least about 75%, 80%, 90%, 95%, or perfectly identical to one of SEQ ID NO. 1-141 as set forth in Table 1. The top preferred embodiments against SCN9A include those in Table 1 labeled as follows: “4/11” (SEQ ID NO: 11); “4/34” (SEQ ID NO: 27); “4/72” (SEQ ID NO: 166); “4/77” (SEQ ID NO: 167); “4/84” (SEQ ID NO: 168); “4/86” (SEQ ID NO: 169); “4/91” (SEQ ID NO: 170); and “4/95” (SEQ ID

NO: 171). Additional preferred embodiments against SCN9A include those in Table 1 labeled as follows: “4/6” (SEQ ID NO: 6); and “4/10” (SEQ ID NO: 10). The preferred candidates show robust and significant knockdown activity (>75%) of Nav1.7 in a dose-dependent manner.

[0069] The top preferred embodiments against SCN10A include those in Table 1 labeled as: “5/62” (SEQ ID NO: 172); “5/65” (SEQ ID NO: 173); “5/81” (SEQ ID NO: 174); “5/84” (SEQ ID NO: 175); “5/88” (SEQ ID NO: 176); and “5/108” (SEQ ID NO: 177). Additional preferred embodiments against SCN10A include those in Table 1 labeled as: “5/8” (SEQ ID NO: 61); “5/18” (SEQ ID NO: 71); “5/20” (SEQ ID NO: 73). These preferred candidates show robust

and significant knockdown activity (>65%) of Nav 1.8 in a dose-dependent manner.

[0070] The top preferred embodiments against both SCN10A and SCN9A include those in Table 1 labeled as: “4/64” (SEQ ID NO: 145) and “5/53” (SEQ ID NO: 157). These preferred embodiments show robust knockdown activity of Nav1.7 and Nav1.8 in a dose-dependent manner.

[0071] The top preferred embodiments against SCN11A include those in Table 1 labeled: “6/1” (SEQ ID NO: 102); “6/3” (SEQ ID NO: 104); and “6/6” (SEQ ID NO: 107). Those three candidates show robust and significant knockdown activity (>70%) of Nav 1.9 in a dose-dependent manner.

TABLE 1

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intrinsic)
4/1	GCCAGTTCCACGGGTCACGA	(SEQ ID NO: 1)	SCN9A
4/2	ATCCAGCCAGTTCACGGGT	(SEQ ID NO: 2)	SCN9A
4/3	CAGGTGTACCCCTCTGGACA	(SEQ ID NO: 3)	SCN9A
4/4	AGCAGCGCAGCGTCTGTGGT	(SEQ ID NO: 4)	SCN9A
4/5	TGCCAGCAGCAGCAGCGTC	(SEQ ID NO: 5)	SCN9A
4/6	TTGCCAGCAGCAGCAGCGT	(SEQ ID NO: 6)	SCN9A
4/7	TTTGCCAGCAGCAGCAGCG	(SEQ ID NO: 7)	SCN9A
4/8	GTTTTGCCAGCAGCAGCAG	(SEQ ID NO: 8)	SCN9A
4/9	GTTTGCCCTGTTCTGTTCTT	(SEQ ID NO: 9)	SCN9A
4/10	TGTGCTCGCCTATGCCCTTC	(SEQ ID NO: 10)	SCN9A
4/11	GTTCTGCTGCTTCGCCTTGC	(SEQ ID NO: 11)	SCN9A
4/12	CCCCCTCTGCTCTCATTGTC	(SEQ ID NO: 12)	SCN9A
4/13	GCCCCCTCTGCTCTCATTGT	(SEQ ID NO: 13)	SCN9A
4/14	GAGCCCCCTCTGCTCTCATT	(SEQ ID NO: 14)	SCN9A
4/15	GTGAGCCCCCTCTGCTCTCA	(SEQ ID NO: 15)	SCN9A
4/16	AGTGAGCCCCCTCTGCTCTC	(SEQ ID NO: 16)	SCN9A
4/17	ACTGCTGCGTCGCTCCTGGG	(SEQ ID NO: 17)	SCN9A
4/18	GCATTTTCCCGTTACCGGC	(SEQ ID NO: 18)	SCN9A
4/19	TGCAGTCCACAGCACTGTGC	(SEQ ID NO: 19)	SCN9A
4/20	GCATGAGGGCTGAGCGTCCA	(SEQ ID NO: 20)	SCN9A
4/21	CTCTCAGGGCTGCTTCTTTT	(SEQ ID NO: 21)	SCN9A
4/22	CTGTTTGCCAGCTTCCAAGT	(SEQ ID NO: 22)	SCN9A
4/23	TTGGTCCAGTCCGGTGGGT	(SEQ ID NO: 23)	SCN9A
4/27	TTGCCTCAGCTTCTTCTTGC	(SEQ ID NO: 24)	SCN9A
4/31	GTTGAGTCCACAGCACTGT	(SEQ ID NO: 25)	SCN9A
4/33	AGGTTACCTAGAGCCCTAC	(SEQ ID NO: 26)	SCN9A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
4/34	GGTTGTTTGCATCAGGTCT	(SEQ ID NO: 27)	SCN9A
4/35	AGGTTACAGCTCTGCTTCTT	(SEQ ID NO: 28)	SCN9A
4/36	CTTGGACCCAGCTTTTTC	(SEQ ID NO: 29)	SCN9A
4/37	GGGTACACAGTCTCCTTC	(SEQ ID NO: 30)	SCN9A
4/38	GAATCCATCTCCCCACTCTC	(SEQ ID NO: 31)	SCN9A
4/39	GCTGCCACCTTTCTTAGGA	(SEQ ID NO: 32)	SCN9A
4/40	GCCAATTCCTGGCCATCCT	(SEQ ID NO: 33)	SCN9A
4/41	CCTTGGATCTCTGCCAGGT	(SEQ ID NO: 34)	SCN9A (i)
4/42	GTCCCTGGAGTCTGTCTGA	(SEQ ID NO: 35)	SCN9A (i)
4/43	CTCCCATATCTCCAGTCTGC	(SEQ ID NO: 36)	SCN9A (i)
4/44	GCTCTTGCTCTGGTTCAGCT	(SEQ ID NO: 37)	SCN9A (i)
4/45	GGTCTTCCAGCTTCTCTGC	(SEQ ID NO: 38)	SCN9A (i)
4/46	CATGTCCCTGTCCATCCCTA	(SEQ ID NO: 39)	SCN9A (i)
4/47	GTGTGGCAGCAGTGACCAGT	(SEQ ID NO: 40)	SCN9A (i)
4/48	CCCTTGCTGGGTCTATGT	(SEQ ID NO: 41)	SCN9A (i)
4/49	GCTCTACCTAGCTGTCAGG	(SEQ ID NO: 42)	SCN9A (i)
4/50	GCTCCTCCTCAGAGTTTGC	(SEQ ID NO: 43)	SCN9A (i)
4/51	GAGCCTCTTCTCTTCAGGCC	(SEQ ID NO: 44)	SCN9A (i)
4/52	TGGCTCATCCAGGCTCATCA	(SEQ ID NO: 45)	SCN9A (i)
4/53	GCATTATTTCCACCAAGTCC	(SEQ ID NO: 46)	SCN9A (i)
4/54	TCTCTTCAGTCTCCTCCACA	(SEQ ID NO: 47)	SCN9A (i)
4/55	CCAGCAGTTGGCAGAGGTTC	(SEQ ID NO: 48)	SCN9A (i)
4/56	GCTTGTGCATCCAGTGCCT	(SEQ ID NO: 49)	SCN9A (i)
4/57	CCTCCATCTGATTCTCCTC	(SEQ ID NO: 50)	SCN9A (i)
4/58	GCCTTCTCACCAGTGCTGCT	(SEQ ID NO: 51)	SCN9A (i)
4/59	TCTAGCCTTCTCACCAGTGC	(SEQ ID NO: 52)	SCN9A (i)
4/60	GGTGGCAGGTCAAGCAGGGT	(SEQ ID NO: 53)	SCN9A (i)
5/1	TCTTGGTCCTTCTGCTCCCT	(SEQ ID NO: 54)	SCN10A
5/2	GCTCCCCGATCAGTTCTGCT	(SEQ ID NO: 55)	SCN10A
5/3	TGTCCGGTGTGTGCTGTAGA	(SEQ ID NO: 56)	SCN10A
5/4	TGGTCTCCTTTGTTTCAGC	(SEQ ID NO: 57)	SCN10A
5/5	GTTGCAGCCCCACCAAGGCA	(SEQ ID NO: 58)	SCN10A
5/6	GGGTCTGCTGGTAGAGGCGT	(SEQ ID NO: 59)	SCN10A
5/7	GCACCTCCTGCTCCTTCCGG	(SEQ ID NO: 60)	SCN10A
5/8	GTTGTCTTCTGTGGAGCCCT	(SEQ ID NO: 61)	SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
5/9	GCTGGTCAAGCAGGGTGGGC	(SEQ ID NO: 62)	SCN10A
5/10	GGCCACGCCCAGCTCTAGCA	(SEQ ID NO: 63)	SCN10A
5/11	GGTGAGGTTCCCCAGTGCCC	(SEQ ID NO: 64)	SCN10A
5/12	GTCGTGCATGTGCCAGCGGG	(SEQ ID NO: 65)	SCN10A
5/13	TCCGTGCCAGGGCCACCTGC	(SEQ ID NO: 66)	SCN10A
5/14	GGGAGCTTGGAGCCCTCCAG	(SEQ ID NO: 67)	SCN10A
5/15	GTTCCAGTGCCTGGGCTCCT	(SEQ ID NO: 68)	SCN10A
5/16	CCAGCTCAGGGATCTTCCTC	(SEQ ID NO: 69)	SCN10A
5/17	GTCTTGCGCACCTGCCAGCC	(SEQ ID NO: 70)	SCN10A
5/18	CGTGGGCTTCTGGTCCAGGT	(SEQ ID NO: 71)	SCN10A
5/19	GCCCGCAGTGGCCGAGAGC	(SEQ ID NO: 72)	SCN10A
5/20	GGGCATCCACCACCCCGC	(SEQ ID NO: 73)	SCN10A
5/21	GGCGCCACCAAGGCATCCA	(SEQ ID NO: 74)	SCN10A
5/22	TGTCCTCCCACTTGGGTTGC	(SEQ ID NO: 75)	SCN10A
5/23	TCCCCCTCTGGTGCCATTGCT	(SEQ ID NO: 76)	SCN10A
5/24	GGGCTCCACAGTCCCTCT	(SEQ ID NO: 77)	SCN10A
5/25	GCAGCCTCCTCCTCAGCTCT	(SEQ ID NO: 78)	SCN10A
5/26	GCCATATCCTCACCCTCTCA	(SEQ ID NO: 79)	SCN10A (i)
5/27	GGGACTGCTTTCTCCCTTCC	(SEQ ID NO: 80)	SCN10A (i)
5/28	TGCCTTGTCTCTGGCCTCCC	(SEQ ID NO: 81)	SCN10A (i)
5/29	GCTTGTTTCCAGTCTCAGC	(SEQ ID NO: 82)	SCN10A (i)
5/30	GACCTTCCTCCCACAGTGCC	(SEQ ID NO: 83)	SCN10A (i)
5/31	GAGCCACCCTCCCACACAGC	(SEQ ID NO: 84)	SCN10A (i)
5/32	ACAGCAGTGTCTCCTTGGCC	(SEQ ID NO: 85)	SCN10A (i)
5/33	CCCAGTGTCCACATGTCTCC	(SEQ ID NO: 86)	SCN10A (i)
5/34	TCTGTTGCTCCCACCAGCTT	(SEQ ID NO: 87)	SCN10A (i)
5/35	GCCTCTTCTGTGGAGGTGGG	(SEQ ID NO: 88)	SCN10A (i)
5/36	CCACTCACCACAGGTTCCC	(SEQ ID NO: 89)	SCN10A (i)
5/37	GGTCTCCTCTGCATTTCCCT	(SEQ ID NO: 90)	SCN10A (i)
5/38	GCCCTGCATGTTCTTGAGGC	(SEQ ID NO: 91)	SCN10A (i)
5/39	GCTGGTGTCCAACCTCTCC	(SEQ ID NO: 92)	SCN10A (i)
5/40	CCAGCCTCTACCAGCCCACT	(SEQ ID NO: 93)	SCN10A (i)
5/41	GCCCTCCCTCTTATCTTACC	(SEQ ID NO: 94)	SCN10A (i)
5/42	GCCACCCTAGTTTTCTCTCC	(SEQ ID NO: 95)	SCN10A (i)
5/43	GGTGCCAGCCTGTTTCAGTCC	(SEQ ID NO: 96)	SCN10A (i)
5/44	GTCCACCAAGCCACCTCC	(SEQ ID NO: 97)	SCN10A (i)

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
5/45	GGAAGCTCCCTGCCAGCCTC	(SEQ ID NO: 98)	SCN10A (i)
5/46	GTCTGGGTCTGGTGGCTGT	(SEQ ID NO: 99)	SCN10A (i)
5/47	GCCCTGCCAGTCACACTGCC	(SEQ ID NO: 100)	SCN10A (i)
5/48	GGCTGATCCTTGCCCTTCTGC	(SEQ ID NO: 101)	SCN10A (i)
6/1	TTGCTCTAGGAGCTGTGGCT	(SEQ ID NO: 102)	SCN11A
6/2	AGCACTCAGTGCTCTCTGCC	(SEQ ID NO: 103)	SCN11A
6/3	GATGGTGATGGCCAGCTCAG	(SEQ ID NO: 104)	SCN11A
6/4	GGCCTCCATCTTGTGATGCT	(SEQ ID NO: 105)	SCN11A
6/5	AGGGCTCCGACAGAGTTGCC	(SEQ ID NO: 106)	SCN11A
6/6	GTCAGGCTTCCAAGGCTCC	(SEQ ID NO: 107)	SCN11A
6/7	AGTCAGGCTTCCAAGGCTC	(SEQ ID NO: 108)	SCN11A
6/8	GGACCACAGTCAGGCTTCCA	(SEQ ID NO: 109)	SCN11A
6/9	GTCGGGCTGTGCGGTTACA	(SEQ ID NO: 110)	SCN11A
6/10	TGTCGGGCTGTGCGGTTAC	(SEQ ID NO: 111)	SCN11A
6/11	CTGTGCGGCTGTGCGGTTA	(SEQ ID NO: 112)	SCN11A
6/12	ACTGTGCGGCTGTGCGGTT	(SEQ ID NO: 113)	SCN11A
6/13	GACTGTGCGGCTGTGCGGT	(SEQ ID NO: 114)	SCN11A
6/14	AGACTGTGCGGCTGTGCGG	(SEQ ID NO: 115)	SCN11A
6/15	CCCCATGTGCCAGTGCCGTA	(SEQ ID NO: 116)	SCN11A
6/16	CCTGGGTCTCTGAGCCCCTT	(SEQ ID NO: 117)	SCN11A
6/17	TCCTGGGTCTCTGAGCCCCT	(SEQ ID NO: 118)	SCN11A
6/18	AAGCTCCTCTGGGTCTCTG	(SEQ ID NO: 119)	SCN11A
6/19	GTGGGCTTCTTGTCTCCTG	(SEQ ID NO: 120)	SCN11A
6/20	GTAGCAGGTTTTCCGCAGGT	(SEQ ID NO: 121)	SCN11A
6/21	GGTACTAGCTCCTCTGCCT	(SEQ ID NO: 122)	SCN11A (i)
6/22	CATCCACCTCCAGACCTCCC	(SEQ ID NO: 123)	SCN11A (i)
6/23	GCCCAAGTCCCTCAAGCCTT	(SEQ ID NO: 124)	SCN11A (i)
6/24	GGTTCAGGTTCCACCCAGC	(SEQ ID NO: 125)	SCN11A (i)
6/25	CTGTCTCCTCCATAGTCTCT	(SEQ ID NO: 126)	SCN11A (i)
6/26	TTCTCCCTGCCTTATGGGT	(SEQ ID NO: 127)	SCN11A (i)
6/27	GCTCCTCCTTGCTTCAGGCT	(SEQ ID NO: 128)	SCN11A (i)
6/28	GCATCCAGGCATCTCAGTGC	(SEQ ID NO: 129)	SCN11A (i)
6/29	GCCCTATGCCTGCCTCAGTG	(SEQ ID NO: 130)	SCN11A (i)
6/30	CCACACCTGTCTGCCTGTGT	(SEQ ID NO: 131)	SCN11A (i)
6/31	GTGTCTCTGCCTCTCTACT	(SEQ ID NO: 132)	SCN11A (i)

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
6/32	CCTGCCTTCTCAGAGTGCCA	(SEQ ID NO: 133)	SCN11A (i)
6/33	GCCCTCTTCTCACCAGACC	(SEQ ID NO: 134)	SCN11A (i)
6/34	CCCATGTCCCTACCTCCTTT	(SEQ ID NO: 135)	SCN11A (i)
6/35	GTCCCATCCCAAGCTAGCC	(SEQ ID NO: 136)	SCN11A (i)
6/36	TCTTTAGGTCTGTGGCCCT	(SEQ ID NO: 137)	SCN11A (i)
6/37	CCCACTCCTCCCTTCTTTGA	(SEQ ID NO: 138)	SCN11A (i)
6/38	GGGCTCTCTTCACTCTGCCT	(SEQ ID NO: 139)	SCN11A (i)
6/39	GGGTCCTCTCTGTGGCCACT	(SEQ ID NO: 140)	SCN11A (i)
6/40	CTCCCTAGCCCTGCCTCTTC	(SEQ ID NO: 141)	SCN11A (i)
4/61	AAAATCCAGCCAGTTCCAC	(SEQ ID NO: 142)	SCN9A/SCN10A
4/62	CAAAATCCAGCCAGTTCCA	(SEQ ID NO: 143)	SCN9A/SCN10A
4/63	TGCAATGTACATGTTCAAC	(SEQ ID NO: 144)	SCN9A/SCN10A
4/64	CTGCAATGTACATGTTAC	(SEQ ID NO: 145)	SCN9A/SCN10A
4/65	ACTGCAATGTACATGTTCA	(SEQ ID NO: 146)	SCN9A/SCN10A
4/66	TGACTGCAATGTACATGTT	(SEQ ID NO: 147)	SCN9A/SCN10A
4/67	ATGACTGCAATGTACATGT	(SEQ ID NO: 148)	SCN9A/SCN10A
4/68	GTCATTTTGGCCATGTTAT	(SEQ ID NO: 149)	SCN9A/SCN10A
4/69	TCAAATAACCCAGAAGCCT	(SEQ ID NO: 150)	SCN9A/SCN10A
4/70	TTCAAATAACCCAGAAGCC	(SEQ ID NO: 151)	SCN9A/SCN10A
4/71	TTTCAAATAACCCAGAAGC	(SEQ ID NO: 152)	SCN9A/SCN10A
5/49	AAAATCCAGCCAGTTCCAA	(SEQ ID NO: 153)	SCN9A/SCN10A
5/50	AGGCCTGGGATCACAGAAA	(SEQ ID NO: 154)	SCN9A/SCN10A
5/51	CAGGCCTGGGATCACAGAA	(SEQ ID NO: 155)	SCN9A/SCN10A
5/52	TCAGGCCTGGGATCACAGA	(SEQ ID NO: 156)	SCN9A/SCN10A
5/53	TTCAGGCCTGGGATCACAG	(SEQ ID NO: 157)	SCN9A/SCN10A
5/54	CTTCAGGCCTGGGATCACAA	(SEQ ID NO: 158)	SCN9A/SCN10A
5/55	TGCTCTGTGAATAAATGCT	(SEQ ID NO: 159)	SCN9A/SCN10A
5/56	TAATTTGGCATCTGTCTTT	(SEQ ID NO: 160)	SCN9A/SCN10A
5/57	TCAGATAACCCAGAAGCCT	(SEQ ID NO: 161)	SCN9A/SCN10A
5/58	TTCAGATAACCCAGAAGCC	(SEQ ID NO: 162)	SCN9A/SCN10A
5/59	TTTCAGATAACCCAGAAGC	(SEQ ID NO: 163)	SCN9A/SCN10A
5/60	GGATGATGAATAGATGGAA	(SEQ ID NO: 164)	SCN9A/SCN10A
4/72	ATGCTTCAGGTTTCCCATGA	SEQ ID NO: 166	SCN9A
4/77	GCTCAGTTTGGATGTTTCAG	SEQ ID NO: 167	SCN9A
4/84	GAGGTTGTTTGTCATCAGGGT	SEQ ID NO: 168	SCN9A
4/86	GACCTTTGCTCATTTTCAGCA	SEQ ID NO: 169	SCN9A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
4/91	GGTTTTCCTGATGTTCCACC	SEQ ID NO: 170	SCN9A
4/95	CATATTGGGCTGCTTGTCT	SEQ ID NO: 171	SCN9A
5/62	ATGGTCCTCCCTTTGTTTCTAG	SEQ ID NO: 172	SCN10A
5/65	CCCTTGAAGAGTTGCAGCCC	SEQ ID NO: 173	SCN10A
5/81	GAACTGGTTGATTTTGCCCA	SEQ ID NO: 174	SCN10A
5/84	GGAGAGTGGTTGCTATTGGT	SEQ ID NO: 175	SCN10A
5/88	CCACTTCTCGAGCTTATTG	SEQ ID NO: 176	SCN10A
5/108	GCCAGCCTCATTTAGCAAAA	SEQ ID NO: 177	SCN10A
4/73	TCTTCTTCACTCTCTAGGGT	SEQ ID NO: 178	SCN9A
4/76	GCCCATAATTCTGCTTCTCC	SEQ ID NO: 179	SCN9A
4/78	GAATGCTGAGTGGTGACTGA	SEQ ID NO: 180	SCN9A
4/79	GGGACCTACTGGCTTGGCTG	SEQ ID NO: 181	SCN9A
4/83	TTTAGACCACCAGGTTTCC	SEQ ID NO: 182	SCN9A
4/85	GGAGGTGTGTTTGATCAGGG	SEQ ID NO: 183	SCN9A
4/89	TCATAGGTTGAGCCTCTGCT	SEQ ID NO: 184	SCN9A
4/94	GGAATTGCTCCTATGAGTGC	SEQ ID NO: 185	SCN9A
4/107	GACTGGTTTTTCTGATGCC	SEQ ID NO: 186	SCN9A
4/110	GGTCAAAGCTGGTTAGGGCC	SEQ ID NO: 187	SCN9A
4/111	AGGTTCACTGCCTGTACTGC	SEQ ID NO: 188	SCN9A
4/117	TGACCTCTAGGACTGAGCA	SEQ ID NO: 189	SCN9A
4/118	GGATTCTCTACTTAACC	SEQ ID NO: 190	SCN9A
4/120	CAGATTTCTCAGCCAGGTCC	SEQ ID NO: 191	SCN9A
5/67	CCATGGTGACTACAGCCAAG	SEQ ID NO: 192	SCN10A
5/71	GGAGCTGCTTGCCAACCAGA	SEQ ID NO: 193	SCN10A
5/73	CTGCAGGTGTTCACCTCCC	SEQ ID NO: 194	SCN10A
5/79	TTGACCCAGAAGAAGCTGCC	SEQ ID NO: 195	SCN10A
5/82	CTCTGGTGCCATTGCTGTTG	SEQ ID NO: 196	SCN10A
5/85	GAGTGATGCAGCCTCCTCCT	SEQ ID NO: 197	SCN10A
5/90	GGGTTAGGCAGAGGGCAGTA	SEQ ID NO: 198	SCN10A
5/107	GTCTTACCCTTCTACCATG	SEQ ID NO: 199	SCN10A
5/109	TTGGTGGGTGTGTTAGGGA	SEQ ID NO: 200	SCN10A
4/126	TGTTTGCCTGGTTCTGTTT	SEQ ID NO: 201	SCN9A/SCN10A
4/127	TCAATGTTTGCCTGGTTCTG	SEQ ID NO: 202	SCN9A/SCN10A
4/128	AAAATCCAGCCAGTTCCACG	SEQ ID NO: 203	SCN9A/SCN10A
4/129	TTGTTTTAAGCCCTCAG	SEQ ID NO: 204	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
4/130	CTTGTTTTAAGCCCTCA	SEQ ID NO: 205	SCN9A/SCN10A
4/131	TTGACAAGATGCACCTG	SEQ ID NO: 206	SCN9A/SCN10A
4/132	CTTGTTTTAAGCCCTCAG	SEQ ID NO: 207	SCN9A/SCN10A
4/133	CCAGAAGTTATTTTCAGA	SEQ ID NO: 208	SCN9A/SCN10A
4/134	CTTGCCACAAGTGATC	SEQ ID NO: 209	SCN9A/SCN10A
4/135	TCTTGCCACAAGTGAT	SEQ ID NO: 210	SCN9A/SCN10A
4/136	AATGTCACCTTGCCAA	SEQ ID NO: 211	SCN9A/SCN10A
4/137	CAATGTCACCTTGCCAA	SEQ ID NO: 212	SCN9A/SCN10A
4/138	GCAGTGATCTTTAGCC	SEQ ID NO: 213	SCN9A/SCN10A
4/139	TCAATGTCACCTTGGCC	SEQ ID NO: 214	SCN9A/SCN10A
4/140	AGCAGTGATCTTTAGC	SEQ ID NO: 215	SCN9A/SCN10A
4/141	ATCAATGTCACCTTGGC	SEQ ID NO: 216	SCN9A/SCN10A
4/142	TTGCATCTCTGCAGCA	SEQ ID NO: 217	SCN9A/SCN10A
4/143	TTTGCTGGCATATCTGA	SEQ ID NO: 218	SCN9A/SCN10A
4/144	CAATGTCACCTTGCCAA	SEQ ID NO: 219	SCN9A/SCN10A
4/145	TCAATGTCACCTTGCCAA	SEQ ID NO: 220	SCN9A/SCN10A
4/146	AGCAGTGATCTTTAGCC	SEQ ID NO: 221	SCN9A/SCN10A
4/147	ATCAATGTCACCTTGCC	SEQ ID NO: 222	SCN9A/SCN10A
4/148	ATCAATGTCACCTTGCCAA	SEQ ID NO: 223	SCN9A/SCN10A
4/149	GTCTCAGTGGCCTCTAT	SEQ ID NO: 224	SCN9A/SCN10A
4/150	AGCACTGTCTCAAAGTC	SEQ ID NO: 225	SCN9A/SCN10A
4/151	CTCATCACTGTATCCCT	SEQ ID NO: 226	SCN9A/SCN10A
4/152	CCAGGTGCAGTGGCTCA	SEQ ID NO: 227	SCN9A/SCN10A
4/153	GACAAGTTCAGGAGCTA	SEQ ID NO: 228	SCN9A/SCN10A
4/154	GCCAGGTGCAGTGGCTC	SEQ ID NO: 229	SCN9A/SCN10A
4/155	GGACAAGTTCAGGAGCT	SEQ ID NO: 230	SCN9A/SCN10A
4/156	AGGACAAGTTCAGGAGC	SEQ ID NO: 231	SCN9A/SCN10A
4/157	TAGGACAAGTTCAGGAG	SEQ ID NO: 232	SCN9A/SCN10A
4/158	GCCAGGTGCAGTGGCTCA	SEQ ID NO: 233	SCN9A/SCN10A
4/159	AGGACAAGTTCAGGAGCT	SEQ ID NO: 234	SCN9A/SCN10A
4/160	TAGGACAAGTTCAGGAGC	SEQ ID NO: 235	SCN9A/SCN10A
4/161	TAGGACAAGTTCAGGAGCT	SEQ ID NO: 236	SCN9A/SCN10A
4/162	AACTCTCATGGAAGCTG	SEQ ID NO: 237	SCN9A/SCN10A
4/163	CAATGGCTGGCTTGCTG	SEQ ID NO: 238	SCN9A/SCN10A
4/164	CTGCTCAGCCTCTGCCT	SEQ ID NO: 239	SCN9A/SCN10A
4/165	AGCCCTTGGGGATGGAG	SEQ ID NO: 240	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
4/166	GTGCCAACGTTTTTCAGT	SEQ ID NO: 241	SCN9A/SCN10A
4/167	ACTTTGTTGCCAGGCTG	SEQ ID NO: 242	SCN9A/SCN10A
4/168	AAATTTTGAGCCAGTCT	SEQ ID NO: 243	SCN9A/SCN10A
4/169	ACAGCAGGCTAAATTTG	SEQ ID NO: 244	SCN9A/SCN10A
4/170	CCTAAACAAGGCTAAGA	SEQ ID NO: 245	SCN9A/SCN10A
4/171	TTGCCAGTAAATTTCTC	SEQ ID NO: 246	SCN9A/SCN10A
4/172	GTTCAATCCAGACATAT	SEQ ID NO: 247	SCN9A/SCN10A
4/173	TTCAAATCCCAGCCTTA	SEQ ID NO: 248	SCN9A/SCN10A
4/174	GTTCAAATCCCAGCCTT	SEQ ID NO: 249	SCN9A/SCN10A
4/175	AGTTCAAATCCCAGCCT	SEQ ID NO: 250	SCN9A/SCN10A
4/176	GTTCAAATCCCAGCCTTA	SEQ ID NO: 251	SCN9A/SCN10A
4/177	AGTTCAAATCCCAGCCTT	SEQ ID NO: 252	SCN9A/SCN10A
4/178	TGTTCCAGCAGCATTATT	SEQ ID NO: 253	SCN9A/SCN10A
4/179	CTGTGTTCCAGTAACATA	SEQ ID NO: 254	SCN9A/SCN10A
4/180	CCTTTATTCCAGGACCTT	SEQ ID NO: 255	SCN9A/SCN10A
4/181	TTTGGGGCAGAATTTTC	SEQ ID NO: 256	SCN9A/SCN10A
4/182	ATGTTGGCCAGGATTTT	SEQ ID NO: 257	SCN9A/SCN10A
4/183	TTCAGTTTCCCAGAAC	SEQ ID NO: 258	SCN9A/SCN10A
4/184	ATTCAGTTTCCCAGAA	SEQ ID NO: 259	SCN9A/SCN10A
4/185	GGTTGCTTGAGCCCAGA	SEQ ID NO: 260	SCN9A/SCN10A
4/186	GCCACAAATAACAGGAT	SEQ ID NO: 261	SCN9A/SCN10A
4/187	TGTGAATAGTGCCACAA	SEQ ID NO: 262	SCN9A/SCN10A
4/188	GTAGGATGATTCTGAAA	SEQ ID NO: 263	SCN9A/SCN10A
4/189	GTGGGATGATTCTGAAA	SEQ ID NO: 264	SCN9A/SCN10A
4/190	AGTAGGATGATTCTGAA	SEQ ID NO: 265	SCN9A/SCN10A
4/191	GGTTCATAGGACACTTT	SEQ ID NO: 266	SCN9A/SCN10A
4/192	GTGATGGGTTTCATAGGA	SEQ ID NO: 267	SCN9A/SCN10A
4/193	TGCTATTGTTTCATGTG	SEQ ID NO: 268	SCN9A/SCN10A
4/194	TGTGATGGGTTTCATAGG	SEQ ID NO: 269	SCN9A/SCN10A
4/195	TTGTGATGGGTTTCATAG	SEQ ID NO: 270	SCN9A/SCN10A
4/196	GTTGTGATGGGTTTCATA	SEQ ID NO: 271	SCN9A/SCN10A
4/197	GAGTGTCTATTGTTTCAT	SEQ ID NO: 272	SCN9A/SCN10A
4/198	GGTTGTGATGGGTTTCAT	SEQ ID NO: 273	SCN9A/SCN10A
4/199	TGGTTGTGATGGGTTCA	SEQ ID NO: 274	SCN9A/SCN10A
4/200	GTGGTTGTGATGGGTTTC	SEQ ID NO: 275	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
4/201	ATGTCTTCAAACCTCCA	SEQ ID NO: 276	SCN9A/SCN10A
4/202	TGTGATGGGTTTCATAGGA	SEQ ID NO: 277	SCN9A/SCN10A
4/203	TTGTGATGGGTTTCATAGG	SEQ ID NO: 278	SCN9A/SCN10A
4/204	GTTGTGATGGGTTTCATAG	SEQ ID NO: 279	SCN9A/SCN10A
4/205	GGTTGTGATGGGTTTCATA	SEQ ID NO: 280	SCN9A/SCN10A
4/206	TGGTTGTGATGGGTTTCAT	SEQ ID NO: 281	SCN9A/SCN10A
4/207	GTGGTTGTGATGGGTTCA	SEQ ID NO: 282	SCN9A/SCN10A
4/208	TTGTGATGGGTTTCATAGGA	SEQ ID NO: 283	SCN9A/SCN10A
4/209	GTTGTGATGGGTTTCATAGG	SEQ ID NO: 284	SCN9A/SCN10A
4/210	GGTTGTGATGGGTTTCATAG	SEQ ID NO: 285	SCN9A/SCN10A
4/211	TGGTTGTGATGGGTTTCATA	SEQ ID NO: 286	SCN9A/SCN10A
4/212	GTGGTTGTGATGGGTTTCAT	SEQ ID NO: 287	SCN9A/SCN10A
4/213	GTTGTGATGGGTTTCATAGGA	SEQ ID NO: 288	SCN9A/SCN10A
4/214	GGTTGTGATGGGTTTCATAGG	SEQ ID NO: 289	SCN9A/SCN10A
4/215	TGGTTGTGATGGGTTTCATAG	SEQ ID NO: 290	SCN9A/SCN10A
4/216	GTGGTTGTGATGGGTTTCATA	SEQ ID NO: 291	SCN9A/SCN10A
4/217	TTGAAGTGTGGTTCATG	SEQ ID NO: 292	SCN9A/SCN10A
4/218	ATTGAAGTGTGGTTCAT	SEQ ID NO: 293	SCN9A/SCN10A
4/219	ATTGAAGTGTGGTTCATG	SEQ ID NO: 294	SCN9A/SCN10A
5/110	TGGTTGCCTGGTTCTGCTC	SEQ ID NO: 295	SCN9A/SCN10A
5/111	GAAGTCGTGCATGTGCCAGC	SEQ ID NO: 296	SCN9A/SCN10A
5/112	GAAGTCGTGCATGTGCCAG	SEQ ID NO: 297	SCN9A/SCN10A
5/113	AGAAGTCGTGCATGTGCCAG	SEQ ID NO: 298	SCN9A/SCN10A
5/114	TGTCAAAGTCGTCTCTCACT	SEQ ID NO: 299	SCN9A/SCN10A
5/115	ATGTCAAAGTCGTCTCTCACT	SEQ ID NO: 300	SCN9A/SCN10A
5/116	TTCAGATAACCCAGAAGCCT	SEQ ID NO: 301	SCN9A/SCN10A
5/117	TTTCAGATAACCCAGAAGCC	SEQ ID NO: 302	SCN9A/SCN10A
5/118	CAGGGTCATCGTCACTT	SEQ ID NO: 303	SCN9A/SCN10A
5/119	GTTGTTTGCATCTTGGA	SEQ ID NO: 304	SCN9A/SCN10A
5/120	GGTTGTTTGCATCTTGG	SEQ ID NO: 305	SCN9A/SCN10A
5/121	TGGTTGTTTGCATCTTG	SEQ ID NO: 306	SCN9A/SCN10A
5/122	TTGAGGATTTTGCATC	SEQ ID NO: 307	SCN9A/SCN10A
5/123	TCTTGGTTGTTTGCATC	SEQ ID NO: 308	SCN9A/SCN10A
5/124	GCCACCTGCAGGTGTT	SEQ ID NO: 309	SCN9A/SCN10A
5/125	CACAAATCTGGAGGTTG	SEQ ID NO: 310	SCN9A/SCN10A
5/126	AGTGCAATGTGGTGGTT	SEQ ID NO: 311	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
5/127	GGTTGTTTGCATCTTGGA	SEQ ID NO: 312	SCN9A/SCN10A
5/128	TGGTTGTTTGCATCTTGG	SEQ ID NO: 313	SCN9A/SCN10A
5/129	GCTCAGGCTCTGCCTTG	SEQ ID NO: 314	SCN9A/SCN10A
5/130	AGCTCAGGCTCTGCCTT	SEQ ID NO: 315	SCN9A/SCN10A
5/131	CAGCTCAGGCTCTGCCT	SEQ ID NO: 316	SCN9A/SCN10A
5/132	CTGCTCAGTGTCTGGCT	SEQ ID NO: 317	SCN9A/SCN10A
5/133	CCAGCTCAGGCTCTGCC	SEQ ID NO: 318	SCN9A/SCN10A
5/134	ACCAGCTCAGGCTCTGC	SEQ ID NO: 319	SCN9A/SCN10A
5/135	TCCAGCTCAGTCTCTGT	SEQ ID NO: 320	SCN9A/SCN10A
5/136	TTCCAGCTCAGTCTCTG	SEQ ID NO: 321	SCN9A/SCN10A
5/137	CAGCTCAGGCTCTGCCTT	SEQ ID NO: 322	SCN9A/SCN10A
5/138	CCAGCTCAGGCTCTGCCT	SEQ ID NO: 323	SCN9A/SCN10A
5/139	ACCAGCTCAGGCTCTGCC	SEQ ID NO: 324	SCN9A/SCN10A
5/140	CCAGCTCAGGCTCTGCCTT	SEQ ID NO: 325	SCN9A/SCN10A
5/141	ACCAGCTCAGGCTCTGCCT	SEQ ID NO: 326	SCN9A/SCN10A
5/142	TTCTGCTCCTGCTGCAG	SEQ ID NO: 327	SCN9A/SCN10A
5/143	CTGCTGTAGTCTAATTC	SEQ ID NO: 328	SCN9A/SCN10A
5/144	GCTGCTGTAGTCTAATT	SEQ ID NO: 329	SCN9A/SCN10A
5/145	TTGAGGTTGCCCTTGAA	SEQ ID NO: 330	SCN9A/SCN10A
5/146	CTTCAGGTGTCTCATGT	SEQ ID NO: 331	SCN9A/SCN10A
5/147	CCTTCAGGTGTCTCATG	SEQ ID NO: 332	SCN9A/SCN10A
5/148	ATTTATGCTTCAAGTCC	SEQ ID NO: 333	SCN9A/SCN10A
5/149	GAAGACATTTATGCAGC	SEQ ID NO: 334	SCN9A/SCN10A
5/150	TTGGTTGTTTCACATGA	SEQ ID NO: 335	SCN9A/SCN10A
5/151	TTTGGTTGTTTCACATG	SEQ ID NO: 336	SCN9A/SCN10A
5/152	GTTTGGTTGTTTCACAT	SEQ ID NO: 337	SCN9A/SCN10A
5/153	GGTTTGGTTGTTTCACA	SEQ ID NO: 338	SCN9A/SCN10A
5/154	TAGGTTTGGTTGTTTCA	SEQ ID NO: 339	SCN9A/SCN10A
5/155	ATCTCACTTTGGATGTA	SEQ ID NO: 340	SCN9A/SCN10A
5/156	CATCTCACTTTGGATGT	SEQ ID NO: 341	SCN9A/SCN10A
5/157	GTTTGGTTGTTTCACATG	SEQ ID NO: 342	SCN9A/SCN10A
5/158	GTTTGGTTGTTTCACATGA	SEQ ID NO: 343	SCN9A/SCN10A
5/159	TGATGAGTGGTAACTGA	SEQ ID NO: 344	SCN9A/SCN10A
5/160	ATGCTTAGAGGTGAGTG	SEQ ID NO: 345	SCN9A/SCN10A
5/161	CTGATGAGTGGTAACTG	SEQ ID NO: 346	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
5/162	ATGTTGAGTGGGACAG	SEQ ID NO: 347	SCN9A/SCN10A
5/163	ACTGATGAGTGGTAACT	SEQ ID NO: 348	SCN9A/SCN10A
5/164	GACTGATGAGTGGTAAC	SEQ ID NO: 349	SCN9A/SCN10A
5/165	ACTGATGAGTGGTAACTG	SEQ ID NO: 350	SCN9A/SCN10A
5/166	GACTGATGAGTGGTAACT	SEQ ID NO: 351	SCN9A/SCN10A
5/167	GACTGATGAGTGGTAACTG	SEQ ID NO: 352	SCN9A/SCN10A
5/168	TTCACCCCTCAGCAATGG	SEQ ID NO: 353	SCN9A/SCN10A
5/169	CAGCATATGTATGCTTA	SEQ ID NO: 354	SCN9A/SCN10A
5/170	TGCTCATTTTGGCAAAT	SEQ ID NO: 355	SCN9A/SCN10A
5/171	TTCTCATTCCTGCAAGT	SEQ ID NO: 356	SCN9A/SCN10A
5/172	TTGCTCATTTTGGCAA	SEQ ID NO: 357	SCN9A/SCN10A
5/173	TTTGCTCATTTTGGCAA	SEQ ID NO: 358	SCN9A/SCN10A
5/174	ATTTGCTCATTTTGGCA	SEQ ID NO: 359	SCN9A/SCN10A
5/175	CTTTGCTCATGGCAGAA	SEQ ID NO: 360	SCN9A/SCN10A
5/176	CCTTTGCTCATGGCAGA	SEQ ID NO: 361	SCN9A/SCN10A
5/177	GCCTTTGCTCATGGCAG	SEQ ID NO: 362	SCN9A/SCN10A
5/178	GGCCCTTGCTCATGGCA	SEQ ID NO: 363	SCN9A/SCN10A
5/179	TGGCCTTTGCTCATGGC	SEQ ID NO: 364	SCN9A/SCN10A
5/180	GTGGCCTTTGCTCATGG	SEQ ID NO: 365	SCN9A/SCN10A
5/181	AGTGGCCTTTGCTCATG	SEQ ID NO: 366	SCN9A/SCN10A
5/182	AAGTGGCCTTTGCTCAT	SEQ ID NO: 367	SCN9A/SCN10A
5/183	AAAGTGGCCTTTGCTCA	SEQ ID NO: 368	SCN9A/SCN10A
5/184	CCTTTGCTCATGGCAGAA	SEQ ID NO: 369	SCN9A/SCN10A
5/185	GGCCCTTGCTCATGGCAG	SEQ ID NO: 370	SCN9A/SCN10A
5/186	TGGCCTTTGCTCATGGCA	SEQ ID NO: 371	SCN9A/SCN10A
5/187	GTGGCCTTTGCTCATGGC	SEQ ID NO: 372	SCN9A/SCN10A
5/188	AAAGTGGCCTTTGCTCAT	SEQ ID NO: 373	SCN9A/SCN10A
5/189	TGGCCTTTGCTCATGGCAG	SEQ ID NO: 374	SCN9A/SCN10A
5/190	GTGGCCTTTGCTCATGGCA	SEQ ID NO: 375	SCN9A/SCN10A
5/191	GTGGCCTTTGCTCATGGCAG	SEQ ID NO: 376	SCN9A/SCN10A
5/192	TTTGTGTCAGGTCTTC	SEQ ID NO: 377	SCN9A/SCN10A
5/193	ATTTGTGTCAGGTCTT	SEQ ID NO: 378	SCN9A/SCN10A
5/194	ACCATCTTGTGAAGGT	SEQ ID NO: 379	SCN9A/SCN10A
5/195	ATTTGTGTCAGGTCTTC	SEQ ID NO: 380	SCN9A/SCN10A
5/196	GTCTGCTCGTATACCTA	SEQ ID NO: 381	SCN9A/SCN10A
5/197	TAATATTGGTGGGCTGT	SEQ ID NO: 382	SCN9A/SCN10A

TABLE 1-continued

Sequences for therapeutic oligonucleotide			
Ref. No. (Target code/ No.)	Sequence	SEQ ID NO	Target ((i) for intronic)
5/198	CTAAAAGTGGTGGCCTC	SEQ ID NO: 383	SCN9A/SCN10A
5/199	ATAATATTGGTGGGCTG	SEQ ID NO: 384	SCN9A/SCN10A
5/200	GCTAAAAGTGGTGGCCT	SEQ ID NO: 385	SCN9A/SCN10A
5/201	AGGGGCAGACTGTGAAG	SEQ ID NO: 386	SCN9A/SCN10A
5/202	TAGGGGCAGACTGTGAA	SEQ ID NO: 387	SCN9A/SCN10A
5/203	AGGGCTAGACAAAGCTG	SEQ ID NO: 388	SCN9A/SCN10A
5/204	GATAGAAATGGTGAAAG	SEQ ID NO: 389	SCN9A/SCN10A
5/205	TAGGGGCAGACTGTGAAG	SEQ ID NO: 390	SCN9A/SCN10A
C/1	GCCAUAAATCCGGTTUCUGC	(SEQ ID NO: 165)	ctrl

TABLE 2

Sequences for Therapeutic Oligonucleotides that bind to hNav1.7 pre-mRNA			
ASO sequence (5' to 3')	SEQ ID NO.	Region	length
TCCTGATGTTCCACCAGATT	SEQ ID NO. 391	exonic	20
GCATTATTCCCACCAGGTC	SEQ ID NO. 392	intronic	19
AGACTATCCATCCAAGTGCT	SEQ ID NO. 393	intronic	20
TGGCCCAAATCATCTACCTC	SEQ ID NO. 394	intronic	20
GCCCACTATTATCCTAGACA	SEQ ID NO. 395	intronic	20
TGCTTCTGCCCTACAATTTC	SEQ ID NO. 396	intronic	20
TCCCATATCTCCAGTCTGCT	SEQ ID NO. 397	intronic	20
GGGTAGATTCACTACATCCA	SEQ ID NO. 398	intronic	20
GTCTCTAGGTTTCCACTCTC	SEQ ID NO. 399	intronic	20
GATCATCACCTTACCAGCT	SEQ ID NO. 400	intronic	20

[0072] As discussed above, a measured level of neural activity at a baseline, under treatment with an irritant, and under treatment with the irritant and a composition comprising an anti-Nav1.7 ASO and an anti-Nav1.8 ASO show that the combination of oligos performs better than either oligo alone. As disclosed herein, the top preferred embodiments against SCN9A (aka Nav1.7) include those with one of SEQ ID NO: 11; SEQ ID NO: 27; and SEQ ID NOs: 166-171. Other preferred embodiments against SCN9A include those with one of SEQ ID NO: 6 and SEQ ID NO: 10. The top preferred embodiments against SCN10A (aka Nav1.8) include those with one of SEQ ID NO: 172; SEQ ID NO: 173; SEQ ID NO: 174; SEQ ID NO: 175; SEQ ID NO: 176; and SEQ ID NO: 177. Other preferred embodiments against SCN10A include those with one of SEQ ID NO: 61; SEQ ID NO: 71; and SEQ ID NO: 73.

[0073] Accordingly, a most preferred combination embodiment of the disclosure includes a composition for

treating pain. The composition includes: a first oligonucleotide that hybridizes to an mRNA encoding a sodium channel protein along a segment of the mRNA that is at least about 90% complementary to one of SEQ ID NO: 6; and SEQ ID NO: 10 or more preferably one of SEQ ID NO: 27; and SEQ ID NOs: 166-171. and a second oligonucleotide that hybridizes to an mRNA encoding a sodium channel protein along a segment of the mRNA that is at least about 90% complementary to one of SEQ ID NO: 61; SEQ ID NO: 71; and SEQ ID NO: 73 or more preferably, one of SEQ ID NO: 172; SEQ ID NO: 173; SEQ ID NO: 174; SEQ ID NO: 175; SEQ ID NO: 176; and SEQ ID NO: 177. In the preferred combination embodiments, each of the therapeutic oligonucleotides may have a gapmer structure that includes a central DNA segment flanked by modified RNA wings.

[0074] Either or both of the wings may include modified RNA bases, e.g., both wings may include 5 consecutive

RNA bases with 2'-O-methoxyethyl ribose modifications. The entirety of each oligonucleotide may be connected via phosphodiester or phosphorothioate linkages or others as will be apparent to the skilled artisan. Preferably the plurality of therapeutic oligonucleotides is provided lyophilized or in solution, for dilution or reconstitution in a clinic for intrathecal injection. That is, packaged in one or more tubes, lyophilized or in solution, are at least thousand to millions of copies of the first oligonucleotide and at least thousand to millions of copies of the second oligonucleotide. Exhibiting the effects as demonstrated by FIG. 11, this preferred combination embodiment of the composition may prove to have unexpected benefits as a non-opioid therapeutic for the treatment of pain.

[0075] Other features and embodiments are within the scope of the disclosure. Embodiments of the disclosure include oligonucleotides, including locked nucleic acid (LNA) antisense oligonucleotides targeting SCN9A or SCN10A which are capable of inhibiting the expression of NaV1.7 or NaV 1.8 in a cell which is expressing NaV1.7 and/or NaV1.8. The oligonucleotide of the invention may be used in the prevention or treatment of pain. The invention further provides advantageous target site sequences on the human NaV1.7 pre-mRNA which may be targeted by oligonucleotide inhibitors of human NaV1.7 such as antisense oligonucleotides or RNAi agents, such as siRNAs or shRNAs.

[0076] The invention provides for an oligonucleotide of 10 to 30 nucleotides in length, which comprises a contiguous nucleotide sequence of 10 to 30 nucleotides in length with at least 90% complementarity, such as 100% complementarity, to a human NaV1.7 target nucleic acid and a human NaV1.8 target nucleic acid, and which is capable of inhibiting the expression of both NaV1.7 or NaV1.8 in a cell. An oligonucleotide **107** may be 100% identical to one of SEQ ID NO 1-101, 142-164, and 166-400, or is at least 90% identical.

[0077] Embodiments include a pharmaceutically acceptable salt of the antisense oligonucleotide according to the invention, or the conjugate according to the invention.

[0078] The invention provides a pharmaceutical composition comprising the antisense oligonucleotide of the invention or the conjugate of the invention and a pharmaceutically acceptable diluent, solvent, carrier, salt and/or adjuvant.

[0079] The invention provides for the antisense oligonucleotide of the invention or the conjugate of the invention or the pharmaceutical salt or composition of the invention for use in medicine.

[0080] The invention provides for the antisense oligonucleotide of the invention or the conjugate of the invention or the pharmaceutical salt or composition of the invention for use in the treatment or prevention or alleviation of pain. The invention provides for the use of the antisense oligonucleotide of the invention or the conjugate of the invention or the pharmaceutical salt or composition of the invention, for the preparation of a medicament for the treatment, prevention or alleviation of pain.

[0081] In some embodiments the pain is chronic pain, neuropathic pain, inflammatory pain, spontaneous pain, or nociceptive pain.

[0082] Oligonucleotides are commonly made in the laboratory by solid-phase chemical synthesis followed by purification and isolation. When referring to a sequence of the oligonucleotide, reference is made to the sequence or order

of nucleobase moieties, or modifications thereof, of the covalently linked nucleotides or nucleosides. The oligonucleotide of the invention may be man-made, i.e., chemically synthesized, and is typically purified or isolated. The oligonucleotide of the invention may comprise one or more modified nucleosides or nucleotides, such as 2' sugar modified nucleosides.

[0083] The modified nucleotides may be independently selected from the group consisting of a deoxy-nucleotide, a 3'-terminal deoxy-thymine (dT) nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-O-allyl-modified nucleotide, 2'-C-alkyl-modified nucleotide, 2'-hydroxyl-modified nucleotide, a 2'-methoxyethyl modified nucleotide, a 2'-O-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, a non-natural base comprising nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a phosphorothioate group, a nucleotide comprising a methylphosphonate group, a nucleotide comprising a 5'-phosphate, a nucleotide comprising a 5'-phosphate mimic, a glycol modified nucleotide, and a 2'-O-(N-methylacetamide) modified nucleotide, and combinations thereof.

[0084] The nitrogenous bases of the ASO may be naturally occurring nucleobases such as adenine, guanine, cytosine, thymidine, uracil, xanthine and hypoxanthine, as well as non-naturally occurring variants, such as substituted purine or substituted pyrimidine, such as nucleobases selected from isocytosine, pseudoisocytosine, 5-methyl cytosine, 5-thiozolo-cytosine, 5-propynyl-cytosine, 5-propynyl-uracil, 5-bromouracil 5-thiazolo-uracil, 2-thio-uracil, 2-thio-thymine, inosine, diaminopurine, 6-aminopurine, 2-aminopurine, 2,6-diaminopurine and 2-chloro-6-aminopurine.

[0085] The nucleobase moieties may be indicated by the letter code for each corresponding nucleobase, e.g. A, T, G, C or U, wherein each letter may optionally include modified nucleobases of equivalent function. For example, in the exemplified oligonucleotides, the nucleobase moieties are selected from A, T, G, C, and 5-methyl cytosine. Optionally, for LNA gapmers, 5-methyl cytosine LNA nucleosides may be used.

[0086] An oligonucleotide **107** of the disclosure is capable of down-regulating (inhibiting) the expression of a sodium channel (NaV1.7, 1.8, or 1.9). In some embodiments the antisense oligonucleotide of the invention is capable of modulating the expression of the target by inhibiting or down-regulating it. Preferably, such modulation produces an inhibition of expression of at least 20% compared to the normal expression level of the target, more preferably at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% inhibition compared to the normal expression level of the target.

[0087] An antisense oligonucleotide (ASO) of the disclosure may decrease the level of the target nucleic acid (e.g. via RNase H cleavage), or may decrease the functionality (or alter the functionality) of the target nucleic acid, e.g. via modulation of splicing of a pre-mRNA.

[0088] An oligonucleotide **107** of the disclosure may comprise one or more nucleosides which have a modified sugar moiety, i.e. a modification of the sugar moiety when compared to the ribose sugar moiety found in DNA and

RNA. Numerous nucleosides with modification of the ribose sugar moiety have been made, primarily with the aim of improving certain properties of oligonucleotides, such as affinity and/or nuclease resistance. Such modifications include those where the ribose ring structure is modified, e.g. by replacement with a hexose ring (HNA), or a bicyclic ring, which typically have a bridge between the C2 and C4 carbons on the ribose ring (LNA), or an unlinked ribose ring which typically lacks a bond between the C2 and C3 carbons (e.g. UNA). Modified nucleosides also include nucleosides where the sugar moiety is replaced with a non-sugar moiety, for example in the case of peptide nucleic acids (PNA), or morpholino nucleic acids.

[0089] Sugar modifications also include modifications made via altering the substituent groups on the ribose ring to groups other than hydrogen, or the 2'-OH group naturally found in DNA and RNA nucleosides. Substituents may, for example be introduced at the 2', 3', 4' or 5' positions.

[0090] The oligonucleotide may include one or more Locked Nucleic Acid (LNA) bases. An LNA may include a 2'-modified nucleoside which comprises a biradical linking the C2' and C4' of the ribose sugar ring of said nucleoside (also referred to as a "2'-4' bridge"), which restricts or locks the conformation of the ribose ring. These nucleosides are also termed bridged nucleic acid or bicyclic nucleic acid (BNA) in the literature. The locking of the conformation of the ribose is associated with an enhanced affinity of hybridization (duplex stabilization) when the LNA is incorporated into an oligonucleotide for a complementary RNA or DNA molecule. This can be routinely determined by measuring the melting temperature of the oligonucleotide/complement duplex. Non limiting, exemplary LNA nucleosides are disclosed in WO 99/014226, WO 00/66604, WO 98/039352, WO 2004/046160, WO 00/047599, WO 2007/134181, WO 2010/077578, WO 2010/036698, WO 2007/090071, WO 2009/006478, WO 2011/156202, WO 2008/154401, WO 2009/067647, and WO 2008/150729, all incorporated by reference.

[0091] Pharmaceutically acceptable salts of oligonucleotides of the disclosure include those salts that retain the biological effectiveness and properties of the free bases or free acids, which are not biologically or otherwise undesirable. The salts are formed with inorganic acids such as hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, particularly hydrochloric acid, and organic acids such as acetic acid, propionic acid, glycolic acid, pyruvic acid, oxalic acid, maleic acid, malonic acid, succinic acid, fumaric acid, tartaric acid, citric acid, benzoic acid, cinnamic acid, a sulfonic acid, or salicylic acid. In addition, those salts may be prepared from addition of an inorganic base or an organic base to the free acid. Salts derived from an inorganic base include, but are not limited to, the sodium, potassium, lithium, ammonium, calcium, magnesium salts. Salts derived from organic bases include, but are not limited to salts of primary, secondary, and tertiary amines, substituted amines including naturally occurring substituted amines, cyclic amines and basic ion exchange resins, such as isopropylamine, trimethylamine, diethylamine, triethylamine, tripropylamine, ethanolamine, lysine, arginine, N-ethylpiperidine, piperidine, polyamine resins.

[0092] An oligonucleotide **107** may mediate or promote nuclease mediated degradation of sodium channel pre-mRNA or mRNA transcripts. Nuclease mediated degradation refers to an oligonucleotide capable of mediating deg-

radation of a complementary nucleotide sequence when forming a duplex with such a sequence. In some embodiments, the oligonucleotide may function via nuclease mediated degradation of the target nucleic acid, where the oligonucleotides of the invention are capable of recruiting a nuclease, particularly an endonuclease, preferably endoribonuclease (RNase), such as RNase H. Examples of oligonucleotide designs which operate via nuclease mediated mechanisms are oligonucleotides which typically comprise a region of at least 5 or 6 consecutive DNA nucleosides and are flanked on one side or both sides by affinity enhancing nucleosides, for example gapmers. The RNase H activity of an antisense oligonucleotide **107** refers to its ability to recruit RNase H when in a duplex with a complementary RNA molecule.

[0093] The antisense oligonucleotide **107** of the invention, or contiguous nucleotide sequence thereof, may be a gapmer, also termed gapmer oligonucleotide or gapmer designs. The antisense gapmers are commonly used to inhibit a target nucleic acid via RNase H mediated degradation. A gapmer oligonucleotide comprises at least three distinct structural regions a 5'-flank, a gap and a 3'-flank, F-G-F' in the '5->3' orientation. The "gap" region (G) comprises a stretch of contiguous DNA nucleotides which enable the oligonucleotide to recruit RNase H. The gap region is flanked by a 5' flanking region (F) comprising one or more sugar modified nucleosides, advantageously high affinity sugar modified nucleosides, and by a 3' flanking region (F') comprising one or more sugar modified nucleosides, advantageously high affinity sugar modified nucleosides. The one or more sugar modified nucleosides in region F and F' enhance the affinity of the oligonucleotide for the target nucleic acid (i.e. are affinity enhancing sugar modified nucleosides). In some embodiments, the one or more sugar modified nucleosides in region F and F' are 2' sugar modified nucleosides, such as high affinity 2' sugar modifications, such as independently selected from LNA and 2'-MOE.

[0094] A mixed wing gapmer is an LNA gapmer wherein one or both of region F and F' comprise a 2' substituted nucleoside, such as a 2' substituted nucleoside independently selected from the group consisting of 2'-O-alkyl-RNA units, 2'-O-methyl-RNA, 2'-amino-DNA units, 2'-fluoro-DNA units, 2'-alkoxy-RNA, MOE units, arabino nucleic acid (ANA) units, 2'-fluoro-ANA units, or combinations thereof. In some embodiments wherein at least one of region F and F', or both region F and F' comprise at least one LNA nucleoside, the remaining nucleosides of region F and F' are independently selected from the group consisting of 2'-MOE and LNA. In some embodiments wherein at least one of region F and F', or both region F and F' comprise at least two LNA nucleosides, the remaining nucleosides of region F and F' are independently selected from the group consisting of 2'-MOE and LNA. In some mixed wing embodiments, one or both of region F and F' may further comprise one or more DNA nucleosides. Gapmer designs are discussed in WO 2008/049085 and WO 2012/109395, both incorporated by reference.

[0095] Conjugation of the oligonucleotide **107** to one or more non-nucleotide moieties may improve the pharmacology of the oligonucleotide, e.g. by affecting the activity, cellular distribution, cellular uptake or stability of the oligonucleotide. In some embodiments the conjugate moiety can modify or enhance the pharmacokinetic properties of the oligonucleotide by improving cellular distribution, bioavail-

ability, metabolism, excretion, permeability, and/or cellular uptake of the oligonucleotide. In particular, the conjugate may target the oligonucleotide to a specific organ, tissue or cell type and thereby enhance the effectiveness of the oligonucleotide in that organ, tissue or cell type. The conjugate may also serve to reduce activity of the oligonucleotide in non-target cell types, tissues or organs, e.g. off target activity or activity in non-target cell types, tissues or organs.

[0096] In an embodiment, the non-nucleotide moiety (conjugate moiety) is selected from the group consisting of carbohydrates, cell surface receptor ligands, drug substances, hormones, lipophilic substances, polymers, proteins, peptides, toxins (e.g. bacterial toxins), vitamins, viral proteins (e.g. capsids) or combinations thereof.

[0097] Oligonucleotides **107** of the disclosure may be provided in pharmaceutical compositions that include any of the aforementioned oligonucleotides and/or oligonucleotide conjugates or salts thereof and a pharmaceutically acceptable diluent, carrier, salt and/or adjuvant. A pharmaceutically acceptable diluent includes ACSF artificial cerebrospinal fluid and pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts. In some embodiments the pharmaceutically acceptable diluent is sterile phosphate buffered saline or sterile sodium carbonate buffer. In some preferred embodiments, diluents for clinical application include Elliott's B solution and/or ACSF artificial cerebrospinal fluid.

[0098] In some embodiments the oligonucleotide of the invention is in the form of a solution in the pharmaceutically acceptable diluent, for example dissolved in PBS or sodium carbonate buffer. The oligonucleotide may be pre-formulated in the solution or in some embodiments may be in the form of a dry powder (e.g. a lyophilized powder) which may be dissolved in the pharmaceutically acceptable diluent prior to administration. Suitably, for example the oligonucleotide may be dissolved in a concentration of 0.1-100 mg/mL, such as 1-10 mg/mL.

[0099] Compositions of the disclosure may be administered to a patient for the prevention or treatment of pain, such as chronic pain, neuropathic pain, inflammatory pain, spontaneous pain, or nociceptive pain. The oligonucleotides of the invention, or the conjugates, salts or pharmaceutical compositions of the invention may be for use as a local analgesic.

[0100] The pain which may be treated by the oligonucleotides of the invention, or the conjugates, salts or pharmaceutical compositions of the invention may be pain wherein the pain signal is in the peripheral nervous system. Indications associated with pain with a significant peripheral component include for example, diabetic neuropathies, cancer, cranial neuralgia, postherpetic neuralgia and post-surgical neuralgia.

[0101] Pain which may be prevented, treated or ameliorated using the oligonucleotide, conjugate, composition or salt of the invention may for example be selected from the group consisting of pain associated with inherited erythromelalgia (IEM), paroxysmal extreme pain disorder (PEPD), trigeminal neuralgia, neuropathic pain, chronic pain, but also general treatment of nociceptive (e.g. compression of a nerve), neuropathic pain (e.g. diabetic neuropathy), visceral pain, cancer pain, or mixed pain. The invention provides for the oligonucleotide, conjugate, composition or salt of the invention for the use for the prevention

or for the treatment of pain, such as chronic pain, neuropathic pain, inflammatory pain, spontaneous pain, cancer pain, or nociceptive pain.

[0102] The disclosure provides methods for treating or preventing pain in a subject, such as a human, who is suffering from or is likely to suffer pain, comprising administering a therapeutically or prophylactically effective amount of an oligonucleotide, an oligonucleotide conjugate or a pharmaceutical composition of the invention to a subject who is suffering from or is susceptible to suffering from pain, such as cancer pain, osteoarthritis pain, chronic pain, neuropathic pain, inflammatory pain, spontaneous pain, or nociceptive pain, where the oligonucleotide is targeted to a sequence complementary to one of SEQ ID NOs: 1-164 and 166-400.

Dual-Knockdown Embodiments

[0103] Embodiments of the disclosure relate to therapeutic compositions for treating pain via multiple targets. The compositions include an oligonucleotide that hybridizes to RNAs from more than one gene for a sodium channel protein (e.g., human Nav1.7/Nav1.8) along a segment of the RNA that is at least about 75% complementary to one of SEQ ID NOs: 142-164 to thereby prevent translation of the RNA into the sodium channel protein. Note: SCN9A is Nav1.7; SCN10A is Nav1.8; and SCN11A is Nav1.9.

[0104] The oligonucleotide may have a sequence that is at least 80% similar to one of SEQ ID NOs: 142-164 (albeit in "gapmer" structure, with a DNA core and RNA wings). The outer wings of the oligonucleotide may include modified RNA chemistry, such as, for example, being made mostly or wholly of 2-methoxyethyl (2'-MOE) RNA bases. Preferred embodiments include a few (e.g., two to four) phosphodiester bonds within the wings with all other inter-base linkages being phosphorothioate. For example, the second, third, fourth, fifteenth, and seventeenth linkages (in the direction written in the sequences) may be phosphodiester, with all others being phosphorothioate. Notably in such embodiments, the outer-most linkages and all linkages involving a DNA base are preferably phosphorothioate, leaving three of the inter-RNA linkages to include either two or three phosphodiester linkages (balance being phosphorothioate). In a preferred embodiment, the composition includes copies of one more oligonucleotides, each with a sequence given by one of SEQ ID NOs: 142-164, with a 5-9-5 (RNA-DNA-RNA) gapmer design, in which the outer-most inter-base linkages and all linkages involving a DNA base are phosphorothioate, and in which the other three inter-RNA linkages of each wing comprise either two or three phosphodiester linkages (balance being phosphorothioate).

[0105] The human Nav1.7/Nav1.8 dual knockdown sequences exemplified by SEQ ID NOs: 142-164 are each preferably provided as a 19-mer ASO gapmer design following a 5×9×5 configuration, with 9-base DNA core with 5' and 3' RNA-like wings. Preferably, the 5' and 3' wings at least substantially follow 2-methoxyethyl (2'-MOE) chemistry and the backbone linkages include a mixture of phosphorothioate (PS) and phosphodiester (PO) (e.g., outer-most inter-base linkages and all linkages involving a DNA base are PS, and in which the other three inter-RNA linkages of each wing comprise either two or three PS linkages, balance PO). In certain embodiments, the human Nav1.7/Nav1.8 dual knockdown comprises an oligonucleotide with a

sequence as given by one of SEQ ID NOs: 142-164 and a gapmer composition as just described.

[0106] Of the dual knockdown sequences, a first preferred embodiment, dubbed group 1, is illustrated by SEQ ID Nos: 142-152, also described as Ref. No. (Target code/No.) 4/61-4/71. All the group 1 dual knockdown ASOs have 100% match with human SCN9A transcript and only have 1 nucleotide mismatch with human SCN10A. The indicated dual-knockdown ASO oligonucleotides of group 1 have a 5-9-5 gapmer design, with a 9-base DNA core and 5' and 3' RNA-like wings. The 5' and 3' wings include 2-methoxyethyl (2'-MOE) chemistry. The backbone includes a mixture of phosphorothioate (PS) and phosphodiester (PO), e.g., in which the second, third, fourth, fifteenth, and seventeenth linkages (in the direction written in the sequences) may be PO, with all others being PS. Preferably any of SEQ ID Nos: 142-152 have that backbone chemistry. Each of the sequences in this group 1 has zero mismatches to a target sequence in human SCN9A pre-RNA or mRNA and 1 mismatch to a target sequence in human SCN10A pre-RNA or mRNA.

[0107] Of the dual knockdown sequences, a second preferred embodiment is dubbed group 2, illustrated by SEQ ID NOs: 153-164, also described as Ref. No. (Target code/No.) 5/49-5/60. All the group 2 dual knockdown ASOs have 100% match with human SCN10A transcript and only have 1 nucleotide mismatch with human SCN9A. The indicated dual-knockdown ASO oligonucleotides of group 2 have a 5-9-5 gapmer design, with 9-base DNA core with 5' and 3' RNA-like wings. The 5' and 3' wings include 2-methoxyethyl (2'-MOE) chemistry. The backbone includes the mixture of phosphorothioate (PS) and phosphodiester (PO), e.g., in which the second, third, fourth, fifteenth, and seventeenth linkages (in the direction written in the sequences) may be PO, with all others being PS. Preferably any of SEQ ID NOs: 153-164 have that backbone chemistry. Each of the sequences in this group 2 has 1 mismatch to a target sequence in human SCN9A pre-RNA or mRNA and zero mismatches to a target sequence in human SCN10A pre-RNA or mRNA.

[0108] In certain dual-knockdown embodiments, the invention provides for an oligonucleotide of 10 to 30 nucleotides in length, which comprises a contiguous nucleotide sequence of 10 to 30 nucleotides in length with at least 90% complementarity, such as 100% complementarity, to a

human Nav1.7 target nucleic acid and a human Nav1.8 target nucleic acid, and which is capable of inhibiting the expression of both Nav1.7 or Nav1.8 in a cell.

[0109] FIG. 12 shows that ASOs of the disclosure provide potent and selective knockdown of multiple Nav targets. In the figure, the top graph shows comparative expression levels of Nav1.7 beside Nav1.5 and Nav1.2, when treated with four different ASOs (labelled ASO1, ASO2, ASO3, and ASO4) at 2 concentrations. The bottom graph shows comparative expression levels of Nav1.8 beside Nav1.5 and Nav1.2, when treated with those same four different ASOs (again, ASO1, ASO2, ASO3, and ASO4) at 2 concentrations. **[0110]** The first group of six bars in the top panel shows that, at either 5 nM or 15 nM concentration of ASO1, expression of Nav1.7 was knocked down significantly compared to expression of Nav1.5 and Nav1.2. Unexpectedly, the first group of six bars in the bottom panel shows that, at either 5 nM or 15 nM concentration of ASO1, expression of Nav1.8 was knocked down significantly compared to expression of Nav1.5 and Nav1.2. That is, the left-most groups of six bars on the top and on the bottom panels show that ASO1 specifically knocks down expression of both Nav1.7 and Nav1.8, compared to Nav1.5 and Nav1.2. Looking at the top and bottom panels together, the second group of parts shows the same thing for ASO2; the third group of bars shows the same result for ASO3; and the fourth group of six bars shows the same result for ASO4. These data strongly show that ASOs of the disclosure are capable of inhibiting the expression of both Nav1.7 or Nav1.8.

[0111] Different Nav channels have different biophysical properties and roles in excitability.

[0112] FIG. 13 shows which Nav channels are substantially or principally involved in excitability at threshold, sub-threshold, and above the threshold for neural activity. The data are obtained by using blue light and Optopatch constructs to stimulate neurons and measure sodium (Na) conductivity. Blue light stimulation protocols were used to characterize roles of each Nav in nociceptor firing. The data show that Nav1.7 and Nav1.8 are primary drivers of activity at and above threshold, and hence that those targets represent valuable targets for a therapeutic composition for treating pain. Here, the disclosure provides antisense oligonucleotide-based therapeutics against validated target Nav1.7 and Nav1.8.

SEQUENCE LISTING

```
Sequence total quantity: 400
SEQ ID NO: 1          moltype = DNA  length = 20
FEATURE              Location/Qualifiers
misc_feature          1..20
                        note = Antisense Oligonucleotide
source                1..20
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 1
gccagttcca cgggtcacga

SEQ ID NO: 2          moltype = DNA  length = 20
FEATURE              Location/Qualifiers
misc_feature          1..20
                        note = Antisense Oligonucleotide
source                1..20
                        mol_type = other DNA
                        organism = synthetic construct
```

-continued

SEQUENCE: 2		
atccagccag ttccacgggt		20
SEQ ID NO: 3	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 3		
caggtgtacc cctctggaca		20
SEQ ID NO: 4	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 4		
agcacgcagc gtctgttggt		20
SEQ ID NO: 5	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 5		
tgccagcagc acgcagcgtc		20
SEQ ID NO: 6	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 6		
ttgccagcag cagcagcgt		20
SEQ ID NO: 7	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 7		
tttgccagca gcacgcagcg		20
SEQ ID NO: 8	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 8		
gttttgccag cagcacgcag		20
SEQ ID NO: 9	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 9		
gtttgcctgg ttctgttctt		20
SEQ ID NO: 10	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	

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source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 10 tgtgctcgcc tatgcccttc		20
SEQ ID NO: 11 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 11 gttctgctgc ttcgccttgc		20
SEQ ID NO: 12 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 12 cccccttctgc tctcattgtc		20
SEQ ID NO: 13 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 13 gcccccttctg ctctcattgt		20
SEQ ID NO: 14 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 14 gagcccccttc tgctctcatt		20
SEQ ID NO: 15 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 15 gtgagccccct tctgctctca		20
SEQ ID NO: 16 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 16 agtgagcccc ttctgctctc		20
SEQ ID NO: 17 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 17 actgctgcgt cgctcctggg		20

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SEQ ID NO: 18	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 18		
gcattttccc gttcaccggc		20
SEQ ID NO: 19	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 19		
tgcaagtccac agcactgtgc		20
SEQ ID NO: 20	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 20		
gcatgagggc tgagcgcca		20
SEQ ID NO: 21	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 21		
ctctcagggc tgcttctttt		20
SEQ ID NO: 22	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 22		
ctgtttgcca gcttccaagt		20
SEQ ID NO: 23	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 23		
ttggtccagt ccggtgggtt		20
SEQ ID NO: 24	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 24		
ttgcctcagc ttcttcttgc		20
SEQ ID NO: 25	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 25 gttcagctcc acagcactgt		20
SEQ ID NO: 26 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 26 aggttaccta gagcccctac		20
SEQ ID NO: 27 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 27 ggttggttgc atcagggtct		20
SEQ ID NO: 28 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 28 aggttcagcc tctgcttctt		20
SEQ ID NO: 29 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 29 cttggaacccc agctttttca		20
SEQ ID NO: 30 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 30 gggttaccac agtctccttc		20
SEQ ID NO: 31 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 31 gaatccatct cccactctc		20
SEQ ID NO: 32 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 32 gctgcccacc tttcttagga		20
SEQ ID NO: 33	moltype = DNA length = 20	

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FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 33		
gccaatcccc tggccatcct		20
SEQ ID NO: 34	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 34		
ccttgggatac tctgccaggt		20
SEQ ID NO: 35	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 35		
gtccctggag tcttgtctga		20
SEQ ID NO: 36	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 36		
ctcccatatc tccagtctgc		20
SEQ ID NO: 37	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 37		
gctcttgctc tggttcagct		20
SEQ ID NO: 38	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 38		
ggttcttcca gcttctctgc		20
SEQ ID NO: 39	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 39		
catgtccctg tccatcccta		20
SEQ ID NO: 40	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 40		
gtgtggcagc agtgaccagt		20
SEQ ID NO: 41	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 41		
cccttgctgtc gggctctatgt		20
SEQ ID NO: 42	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 42		
gctctaccct agctgtcagg		20
SEQ ID NO: 43	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 43		
gtctctctctc agagttttgc		20
SEQ ID NO: 44	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 44		
gagcctcttc tcttcaggcc		20
SEQ ID NO: 45	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 45		
tggctcatcc aggctcatca		20
SEQ ID NO: 46	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 46		
gcattattcc caccaggtcc		20
SEQ ID NO: 47	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 47		
tctcttcagt ctctccaca		20
SEQ ID NO: 48	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	

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source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 48 ccagcagttg gcagaggttc		20
SEQ ID NO: 49 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 49 gcttgtgcat cccagtgcct		20
SEQ ID NO: 50 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 50 cctccatctg attcctcctc		20
SEQ ID NO: 51 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 51 gccttctcac cagtgtgtgt		20
SEQ ID NO: 52 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 52 tctagccttc tcaccagtgc		20
SEQ ID NO: 53 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 53 ggtggcaggt caagcagggt		20
SEQ ID NO: 54 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 54 tcttggctct tctgtccct		20
SEQ ID NO: 55 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 55 gctccccgat cagttctgt		20

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SEQ ID NO: 56	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 56		
tgcccggtgt gtgctgtaga		20
SEQ ID NO: 57	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 57		
tggctctccc ttgttcagc		20
SEQ ID NO: 58	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 58		
gttcagccc caccaaggca		20
SEQ ID NO: 59	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 59		
gggtctgctg gtagaggcgt		20
SEQ ID NO: 60	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 60		
gcacctctcg ctcttccgg		20
SEQ ID NO: 61	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 61		
gttgtcttct gtggagccct		20
SEQ ID NO: 62	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 62		
gctggtcaag cagggtgggc		20
SEQ ID NO: 63	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 63 ggccacgccc agctctagca		20
SEQ ID NO: 64 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 64 ggtgagggttc cccagtgcgc		20
SEQ ID NO: 65 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 65 gtcgtgcatg tgccagcggg		20
SEQ ID NO: 66 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 66 tccgtgccag ggccacctgc		20
SEQ ID NO: 67 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 67 gggagcttgg agccctccag		20
SEQ ID NO: 68 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 68 gttcagtgctg ctgggctcct		20
SEQ ID NO: 69 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 69 ccagctcagg gatcttcctc		20
SEQ ID NO: 70 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 70 gtcttgcgca cctgccagcc		20
SEQ ID NO: 71	moltype = DNA length = 20	

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FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 71		
cgtgggcttc tggtcaggt		20
SEQ ID NO: 72	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 72		
gccccgagtg gccgcagagc		20
SEQ ID NO: 73	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 73		
gggcatccac caccacccgc		20
SEQ ID NO: 74	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 74		
ggcgcccacc agggcatcca		20
SEQ ID NO: 75	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 75		
tgtcctccca cttgggttgc		20
SEQ ID NO: 76	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 76		
tcccctctgg tgccattgct		20
SEQ ID NO: 77	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 77		
gggctccac agtcccctct		20
SEQ ID NO: 78	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 78 gcagcctcct cctcagctct	20
SEQ ID NO: 79 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 79 gccatatacct caccctctca	20
SEQ ID NO: 80 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 80 gggactgctt tctcccttcc	20
SEQ ID NO: 81 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 81 tgccttgctc ctggcctccc	20
SEQ ID NO: 82 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 82 gcttggttcc agtcctcagc	20
SEQ ID NO: 83 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 83 gaccttcctc ccacagtgcc	20
SEQ ID NO: 84 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 84 gagccaccct cccacacagc	20
SEQ ID NO: 85 FEATURE misc_feature source	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct
SEQUENCE: 85 acagcagtgt ctccttgccc	20
SEQ ID NO: 86 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20

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source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 86		
cccagtgtcc acatgtctcc		20
SEQ ID NO: 87	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 87		
tctgttgctc ccaccagctt		20
SEQ ID NO: 88	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 88		
gcctcttctg tggaggtggg		20
SEQ ID NO: 89	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 89		
ccactcacca ccaggttccc		20
SEQ ID NO: 90	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 90		
ggtctcctct gcatttcctt		20
SEQ ID NO: 91	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 91		
gccctgcatg ttctgagggc		20
SEQ ID NO: 92	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 92		
gctggctgtc caacctctcc		20
SEQ ID NO: 93	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 93		
ccagcctcta ccagccact		20

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SEQ ID NO: 94	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 94		
gccctccctc ttatcttacc		20
SEQ ID NO: 95	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 95		
gccaccctag tttctctccc		20
SEQ ID NO: 96	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 96		
ggtgccagcc tgttcagtc		20
SEQ ID NO: 97	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 97		
gtccacccaa gcccacctcc		20
SEQ ID NO: 98	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 98		
ggaactccct gccagcctc		20
SEQ ID NO: 99	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 99		
gtctgggtcc tgggtgctgt		20
SEQ ID NO: 100	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 100		
gccctgccag tcacactgcc		20
SEQ ID NO: 101	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 101 ggctgacct tgccttctgc		20
SEQ ID NO: 102 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 102 ttgctctagg agctgtggct		20
SEQ ID NO: 103 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 103 agcactcagt gctctctgcc		20
SEQ ID NO: 104 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 104 gatggtgatg gccagctcag		20
SEQ ID NO: 105 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 105 ggcctccatc ttgtgatgct		20
SEQ ID NO: 106 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 106 agggtccga cagagttgcc		20
SEQ ID NO: 107 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 107 gtcaggcttc caagggtcc		20
SEQ ID NO: 108 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide 1..20	
source	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 108 agtcaggctt ccaagggtc		20
SEQ ID NO: 109	moltype = DNA length = 20	

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FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 109		
ggaccacagt caggcttcca		20
SEQ ID NO: 110	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 110		
gtcgggcctg tcgggttaca		20
SEQ ID NO: 111	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 111		
tgtcgggcct gtcgggttac		20
SEQ ID NO: 112	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 112		
ctgtcgggcc tgtcgggtta		20
SEQ ID NO: 113	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 113		
actgtcgggc ctgtcgggtt		20
SEQ ID NO: 114	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 114		
gactgtcggg cctgtcgggt		20
SEQ ID NO: 115	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 115		
agactgtcgg gcctgtcggg		20
SEQ ID NO: 116	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 116		
ccccatgtgc cagtgccgta		20
SEQ ID NO: 117	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 117		
cctgggtctc tgagcccctt		20
SEQ ID NO: 118	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 118		
tcctgggtct ctgagcccct		20
SEQ ID NO: 119	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 119		
aagctcctcc tgggtctctg		20
SEQ ID NO: 120	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 120		
gtgggcttct tggtctctctg		20
SEQ ID NO: 121	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 121		
gtagcaggtt ttccgcaggt		20
SEQ ID NO: 122	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 122		
ggtactagct cctcctgctt		20
SEQ ID NO: 123	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 123		
catccacctc cagacctccc		20
SEQ ID NO: 124	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	

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source	note = Antisense Oligonucleotide 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 124 gccccaggtcc ctcaagcctt		20
SEQ ID NO: 125 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 125 ggttccaggt tccaccagc		20
SEQ ID NO: 126 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 126 ctgtctcctc cataggtcct		20
SEQ ID NO: 127 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 127 ttcctccctg ccttatgggt		20
SEQ ID NO: 128 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 128 gctcctcctt gcttcaggct		20
SEQ ID NO: 129 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 129 gcatccaggc atctcagtg		20
SEQ ID NO: 130 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 130 gccctatgcc tgcctcagtg		20
SEQ ID NO: 131 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 131 ccacacctgt ctgcctgtgt		20

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SEQ ID NO: 132	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 132		
gtgtcctctg cctctctact		20
SEQ ID NO: 133	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 133		
cctgccttct cagagtgccca		20
SEQ ID NO: 134	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 134		
gccctctttc tcaccagacc		20
SEQ ID NO: 135	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 135		
cccatgtccc tacctccttt		20
SEQ ID NO: 136	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 136		
gtcccatccc aagtctagcc		20
SEQ ID NO: 137	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 137		
tctttaggtc ctggtgcct		20
SEQ ID NO: 138	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 138		
cccactcctc ccttctttga		20
SEQ ID NO: 139	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	

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mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 139	
gggtctctctt cactctgcct	20
SEQ ID NO: 140	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
note = Antisense Oligonucleotide	
source	1..20
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 140	
gggtcctctc tgttgccact	20
SEQ ID NO: 141	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
note = Antisense Oligonucleotide	
source	1..20
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 141	
ctccctagcc ctgcctcttc	20
SEQ ID NO: 142	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
note = Antisense Oligonucleotide	
source	1..19
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 142	
aaaatccagc cagttccac	19
SEQ ID NO: 143	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
note = Antisense Oligonucleotide	
source	1..19
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 143	
caaaatccag ccagttcca	19
SEQ ID NO: 144	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
note = Antisense Oligonucleotide	
source	1..19
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 144	
tgcaatgtac atgttcacc	19
SEQ ID NO: 145	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
note = Antisense Oligonucleotide	
source	1..19
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 145	
ctgcaatgta catgttcac	19
SEQ ID NO: 146	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
note = Antisense Oligonucleotide	
source	1..19
mol_type = other DNA	
organism = synthetic construct	
SEQUENCE: 146	
actgcaatgt acatgttca	19
SEQ ID NO: 147	moltype = DNA length = 19

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FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 147		
tgactgcaat gtacatggt		19
SEQ ID NO: 148	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 148		
atgactgcaa tgtacatgt		19
SEQ ID NO: 149	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 149		
gtcatttttg ccatgttat		19
SEQ ID NO: 150	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 150		
tcaaataacc cagaagcct		19
SEQ ID NO: 151	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 151		
ttcaaataac ccagaagcc		19
SEQ ID NO: 152	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 152		
tttcaaataa cccagaagc		19
SEQ ID NO: 153	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 153		
aaaatccagc cagttccaa		19
SEQ ID NO: 154	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	

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SEQUENCE: 154 aggcctggga tcacagaaa	19
SEQ ID NO: 155 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 155 caggcctggg atcacagaa	19
SEQ ID NO: 156 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 156 tcaggcctgg gatcacaga	19
SEQ ID NO: 157 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 157 ttcaggcctg ggatcacag	19
SEQ ID NO: 158 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 158 cttcaggcct gggatcaca	19
SEQ ID NO: 159 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 159 tgctctgtga ataaatgct	19
SEQ ID NO: 160 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 160 taatttggca tctgtcttt	19
SEQ ID NO: 161 FEATURE misc_feature source	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct
SEQUENCE: 161 tcagataacc cagaagcct	19
SEQ ID NO: 162 FEATURE misc_feature	moltype = DNA length = 19 Location/Qualifiers 1..19

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source	note = Antisense Oligonucleotide 1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 162 ttcagataac ccagaagcc		19
SEQ ID NO: 163 FEATURE misc_feature	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 163 tttcagataa cccagaagc		19
SEQ ID NO: 164 FEATURE misc_feature	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 164 ggatgatgaa tagatggaa		19
SEQ ID NO: 165 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Control Sequence	
source	1..20 mol_type = other DNA organism = synthetic construct	
modified_base	5 mod_base = OTHER note = uracil	
modified_base	16 mod_base = OTHER note = uracil	
modified_base	18 mod_base = OTHER note = uracil	
SEQUENCE: 165 gccataatcc gggtttctgc		20
SEQ ID NO: 166 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 166 atgcttcagg tttcccatga		20
SEQ ID NO: 167 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 167 gctcagtttg gatgtttcag		20
SEQ ID NO: 168 FEATURE misc_feature	moltype = DNA length = 20 Location/Qualifiers 1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 168 gaggttggtt gcatcagggt		20

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SEQ ID NO: 169	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 169		
gacctttgct catttcagca		20
SEQ ID NO: 170	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 170		
ggttttcctg atgttcacc		20
SEQ ID NO: 171	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 171		
catatttggg ctgcttgct		20
SEQ ID NO: 172	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 172		
atggtcctcc cttgttcag		20
SEQ ID NO: 173	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 173		
cccttgaaga gttgcagccc		20
SEQ ID NO: 174	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 174		
gaactggttg attttgccca		20
SEQ ID NO: 175	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 175		
ggagagtggg tgctattggt		20
SEQ ID NO: 176	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	

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SEQUENCE: 176	organism = synthetic construct	
ccactttctg gagcttattg		20
SEQ ID NO: 177	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 177		
gccagcctca tttagcaaaa		20
SEQ ID NO: 178	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 178		
tcttcttcac tctctagggt		20
SEQ ID NO: 179	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 179		
gcccataatt ctgcttctcc		20
SEQ ID NO: 180	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 180		
gaatgctgag tggtgactga		20
SEQ ID NO: 181	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 181		
gggacctact ggcttggtg		20
SEQ ID NO: 182	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 182		
tttaggacca ccagggttcc		20
SEQ ID NO: 183	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 183		
ggaggttggt tgcacaggg		20
SEQ ID NO: 184	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

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misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 184		
tcataggttc agcctctgct		20
SEQ ID NO: 185	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 185		
ggaattgctc ctatgagtgc		20
SEQ ID NO: 186	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 186		
gactgggtttt tcctgatgcc		20
SEQ ID NO: 187	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 187		
ggtcaaagct ggtagggcc		20
SEQ ID NO: 188	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 188		
aggttcactg cctgtactgc		20
SEQ ID NO: 189	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 189		
tgaccctcta ggactgagca		20
SEQ ID NO: 190	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 190		
ggattcctca cttactaacc		20
SEQ ID NO: 191	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 191		

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cagatttctc agccaggtcc	20
SEQ ID NO: 192	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 192	
ccatggtgac tacagccaag	20
SEQ ID NO: 193	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 193	
ggagctgctt gccaaccca	20
SEQ ID NO: 194	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 194	
ctgcaggttg ttcacctccc	20
SEQ ID NO: 195	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 195	
ttgaccacaga agaagctgcc	20
SEQ ID NO: 196	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 196	
ctctggtgcc attgctgttg	20
SEQ ID NO: 197	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 197	
gagtgatgca gcctcctcct	20
SEQ ID NO: 198	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 198	
gggttaggca gagggcagta	20
SEQ ID NO: 199	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Antisense Oligonucleotide

-continued

source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 199		
gtcttaccct tcctaccatg		20
SEQ ID NO: 200	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 200		
ttggtgggtg tggtaggga		20
SEQ ID NO: 201	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19 note = Antisense Oligonucleotide	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 201		
tgttgcctg gttctgttc		19
SEQ ID NO: 202	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 202		
tcaatgtttg cctggttctg		20
SEQ ID NO: 203	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20 note = Antisense Oligonucleotide	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 203		
aaaatccagc cagttccacg		20
SEQ ID NO: 204	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 204		
ttgttttaag ccctcag		17
SEQ ID NO: 205	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 205		
ctgttttaaa gccctca		17
SEQ ID NO: 206	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 206		
ttgacaagat gcacctg		17

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SEQ ID NO: 207	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 207		
ctgtttttaa gccctcag		18
SEQ ID NO: 208	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 208		
ccagaagtta ttccaga		17
SEQ ID NO: 209	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 209		
cttgccaca agtgatc		17
SEQ ID NO: 210	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 210		
tcttgccac aagtgat		17
SEQ ID NO: 211	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 211		
aatgtcacct tggccaa		17
SEQ ID NO: 212	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 212		
caatgtcacc ttggcca		17
SEQ ID NO: 213	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 213		
gcagtgtatc tttagcc		17
SEQ ID NO: 214	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	

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SEQUENCE: 214	organism = synthetic construct	
tcaatgtcac cttggcc		17
SEQ ID NO: 215	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 215		
agcagtgtat ctttagc		17
SEQ ID NO: 216	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 216		
atcaatgtca ccttgcc		17
SEQ ID NO: 217	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 217		
ttgcatctct gtcagca		17
SEQ ID NO: 218	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 218		
tttgctggca tatctga		17
SEQ ID NO: 219	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 219		
caatgtcacc ttggccaa		18
SEQ ID NO: 220	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 220		
tcaatgtcac cttggcca		18
SEQ ID NO: 221	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 221		
agcagtgtat ctttagcc		18
SEQ ID NO: 222	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	

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misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 222		
atcaatgtca ccttggcc		18
SEQ ID NO: 223	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 223		
atcaatgtca ccttgcca		19
SEQ ID NO: 224	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 224		
gtctcagtg cctctat		17
SEQ ID NO: 225	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 225		
agcactgtct caaagtc		17
SEQ ID NO: 226	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 226		
ctcatcactg tatccct		17
SEQ ID NO: 227	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 227		
ccaggtgcag tggctca		17
SEQ ID NO: 228	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 228		
gacaagttca ggagcta		17
SEQ ID NO: 229	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 229		

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gccaggtgca gtggctc	17
SEQ ID NO: 230	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 230	
ggacaagttc aggagct	17
SEQ ID NO: 231	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 231	
aggacaagtt caggagc	17
SEQ ID NO: 232	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 232	
taggacaagt tcaggag	17
SEQ ID NO: 233	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide
source	1..18
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 233	
gccaggtgca gtggctca	18
SEQ ID NO: 234	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide
source	1..18
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 234	
aggacaagtt caggagct	18
SEQ ID NO: 235	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide
source	1..18
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 235	
taggacaagt tcaggagc	18
SEQ ID NO: 236	moltype = DNA length = 19
FEATURE	Location/Qualifiers
misc_feature	1..19
	note = Antisense Oligonucleotide
source	1..19
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 236	
taggacaagt tcaggagct	19
SEQ ID NO: 237	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide

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source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 237 aactctcatg gaagctg		17
SEQ ID NO: 238 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 238 caatggctgg cttgctg		17
SEQ ID NO: 239 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 239 ctgctcagcc tctgctt		17
SEQ ID NO: 240 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 240 agcccttggg gatggag		17
SEQ ID NO: 241 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 241 gtgccaacgt tttcagt		17
SEQ ID NO: 242 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 242 actttgttgc caggetg		17
SEQ ID NO: 243 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 243 aaattttgag ccagtct		17
SEQ ID NO: 244 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 244 acagcaggct aaaattg		17

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SEQ ID NO: 245	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 245		
cctaacaag gctaaga		17
SEQ ID NO: 246	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 246		
ttgccagtaa aattctc		17
SEQ ID NO: 247	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 247		
gttcattcca gacatat		17
SEQ ID NO: 248	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 248		
ttcaaatccc agcctta		17
SEQ ID NO: 249	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 249		
gttcaaatcc cagcctt		17
SEQ ID NO: 250	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 250		
agttcaaatc ccagcct		17
SEQ ID NO: 251	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 251		
gttcaaatcc cagcctta		18
SEQ ID NO: 252	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	

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SEQUENCE: 252	organism = synthetic construct	
agttcaaata ccagcctt		18
SEQ ID NO: 253	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 253		
tggtcagcag cattatt		17
SEQ ID NO: 254	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 254		
ctgtgttcag taacata		17
SEQ ID NO: 255	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 255		
cctttattca ggacctt		17
SEQ ID NO: 256	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 256		
tttggggcag aattttc		17
SEQ ID NO: 257	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 257		
atgttgcca ggatttt		17
SEQ ID NO: 258	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 258		
ttcagtttgc ccagaac		17
SEQ ID NO: 259	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 259		
attcagtttg cccagaa		17
SEQ ID NO: 260	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	

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misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 260		
ggttgcttga gcccaga		17
SEQ ID NO: 261	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 261		
gccacaaata acaggat		17
SEQ ID NO: 262	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 262		
tgtgaatagt gccacaa		17
SEQ ID NO: 263	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 263		
gtaggatgat tctgaaa		17
SEQ ID NO: 264	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 264		
gtgggatgat tctgaaa		17
SEQ ID NO: 265	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 265		
agtaggatga ttctgaa		17
SEQ ID NO: 266	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 266		
ggttcataagg acacttt		17
SEQ ID NO: 267	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 267		

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gtgatgggtt catagga	17
SEQ ID NO: 268	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 268	
tgctattggt tcatgtg	17
SEQ ID NO: 269	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 269	
tgtgatgggt tcatagg	17
SEQ ID NO: 270	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 270	
ttgtgatggg ttcatag	17
SEQ ID NO: 271	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 271	
gttgtgatgg gttcata	17
SEQ ID NO: 272	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 272	
gagtgctatt gtttcat	17
SEQ ID NO: 273	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 273	
ggttgtgatg gggtcat	17
SEQ ID NO: 274	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 274	
tggttgtgat gggttca	17
SEQ ID NO: 275	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide

-continued

source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 275 gtggttgga tgggttc		17
SEQ ID NO: 276 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 276 atgtcttcaa acttcca		17
SEQ ID NO: 277 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 277 tgtgatgggt tcatagga		18
SEQ ID NO: 278 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 278 ttgtgatggg ttcatagg		18
SEQ ID NO: 279 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 279 gttgtgatgg gttcatag		18
SEQ ID NO: 280 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 280 ggttgtgatg gggtcata		18
SEQ ID NO: 281 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 281 tgggtgtgat gggttcata		18
SEQ ID NO: 282 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 282 gtggttgga tgggttca		18

-continued

SEQ ID NO: 283	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 283		
ttgtgatggg ttcatagga		19
SEQ ID NO: 284	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 284		
gttgtgatgg gttcatagg		19
SEQ ID NO: 285	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 285		
ggttgtgatg ggttcatag		19
SEQ ID NO: 286	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 286		
tggttgtgat gggttcata		19
SEQ ID NO: 287	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 287		
gtggttgtga tgggttcata		19
SEQ ID NO: 288	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 288		
gttgtgatgg gttcatagga		20
SEQ ID NO: 289	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 289		
ggttgtgatg ggttcatagg		20
SEQ ID NO: 290	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	

-continued

SEQUENCE: 290	organism = synthetic construct	
tggttgat ggggtcatag		20
SEQ ID NO: 291	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide	
	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 291		
gtggttgga ggggtcata		20
SEQ ID NO: 292	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 292		
ttgaagtgtt gggtcatg		17
SEQ ID NO: 293	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 293		
attgaagtgt gggtcatg		17
SEQ ID NO: 294	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 294		
attgaagtgt gggtcatg		18
SEQ ID NO: 295	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
source	note = Antisense Oligonucleotide	
	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 295		
tggttgctg gttctgctc		19
SEQ ID NO: 296	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
source	note = Antisense Oligonucleotide	
	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 296		
gaagtcgtgc atgtgccagc		20
SEQ ID NO: 297	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
source	note = Antisense Oligonucleotide	
	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 297		
gaagtcgtgc atgtgccag		19
SEQ ID NO: 298	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

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misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 298		
agaagtcgtg catgtgccag		20
SEQ ID NO: 299	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 299		
tgtcaaagtc gtcctcact		19
SEQ ID NO: 300	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 300		
atgtcaaagtc gtcctcact		20
SEQ ID NO: 301	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 301		
ttcagataac ccagaagcct		20
SEQ ID NO: 302	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 302		
tttcagataa cccagaagcc		20
SEQ ID NO: 303	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 303		
cagggtcatc gtcactt		17
SEQ ID NO: 304	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 304		
gttggttgca tcttgga		17
SEQ ID NO: 305	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 305		

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ggttgtttgc atcttgg	17
SEQ ID NO: 306	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 306	
tggttggttg catcttg	17
SEQ ID NO: 307	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 307	
ttgaggattt ttgcatc	17
SEQ ID NO: 308	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 308	
tcttggttgt ttgcatc	17
SEQ ID NO: 309	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 309	
gccacctgca gggtggt	17
SEQ ID NO: 310	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 310	
cacaaatctg gaggttg	17
SEQ ID NO: 311	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 311	
agtgcaatgt ggtggtt	17
SEQ ID NO: 312	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide
source	1..18
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 312	
ggttgtttgc atcttgga	18
SEQ ID NO: 313	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide

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source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 313 tggttggttg catcttgg		18
SEQ ID NO: 314 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 314 gctcaggctc tgccttg		17
SEQ ID NO: 315 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 315 agctcaggct ctgcctt		17
SEQ ID NO: 316 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 316 cagctcaggc tctgctt		17
SEQ ID NO: 317 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 317 ctgctcagtg tctggct		17
SEQ ID NO: 318 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 318 ccagctcagg ctctgcc		17
SEQ ID NO: 319 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 319 accagctcag gctctgc		17
SEQ ID NO: 320 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 320 tccagctcag tctctgt		17

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SEQ ID NO: 321	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 321		
ttccagctca gtctctg		17
SEQ ID NO: 322	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 322		
cagctcaggc tctgcctt		18
SEQ ID NO: 323	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 323		
ccagctcagg ctctgcct		18
SEQ ID NO: 324	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 324		
accagctcag gctctgcc		18
SEQ ID NO: 325	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 325		
ccagctcagg ctctgcctt		19
SEQ ID NO: 326	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 326		
accagctcag gctctgcct		19
SEQ ID NO: 327	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 327		
ttctgctcct gctgcag		17
SEQ ID NO: 328	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	

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SEQUENCE: 328	organism = synthetic construct	
ctgctgtagt ctaattc		17
SEQ ID NO: 329	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 329		
gctgctgtag tctaatt		17
SEQ ID NO: 330	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 330		
ttgaggttgc ccttgaa		17
SEQ ID NO: 331	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 331		
cttcaggtgt ctcatgt		17
SEQ ID NO: 332	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 332		
ccttcaggtg tctcatg		17
SEQ ID NO: 333	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 333		
atttatgctt caagtcc		17
SEQ ID NO: 334	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 334		
gaagacattt atgcagc		17
SEQ ID NO: 335	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 335		
ttggttggtt cacaatga		17
SEQ ID NO: 336	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	

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misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 336		
tttggttggt tccatg		17
SEQ ID NO: 337	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 337		
gtttggttggt ttcacat		17
SEQ ID NO: 338	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 338		
ggtttggttg tttcaca		17
SEQ ID NO: 339	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 339		
taggttggt tgtttca		17
SEQ ID NO: 340	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 340		
atctcacttt ggatgta		17
SEQ ID NO: 341	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 341		
catctcactt tggatgt		17
SEQ ID NO: 342	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 342		
gtttggttggt ttcacatg		18
SEQ ID NO: 343	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 343		

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gtttggttgt ttcacatga	19
SEQ ID NO: 344	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 344	
tgatgagtgg taactga	17
SEQ ID NO: 345	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 345	
atgcttagag gtgagtg	17
SEQ ID NO: 346	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 346	
ctgatgagtg gtaactg	17
SEQ ID NO: 347	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 347	
atgttgagtg gggacag	17
SEQ ID NO: 348	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 348	
actgatgagt ggtaact	17
SEQ ID NO: 349	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 349	
gactgatgag tggtaac	17
SEQ ID NO: 350	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide
source	1..18
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 350	
actgatgagt ggtaactg	18
SEQ ID NO: 351	moltype = DNA length = 18
FEATURE	Location/Qualifiers
misc_feature	1..18
	note = Antisense Oligonucleotide

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source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 351 gactgatgag tggtaact		18
SEQ ID NO: 352 FEATURE misc_feature	moltype = DNA length = 19 Location/Qualifiers 1..19 note = Antisense Oligonucleotide	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 352 gactgatgag tggtaactg		19
SEQ ID NO: 353 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 353 ttcacctca gcaatgg		17
SEQ ID NO: 354 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 354 cagcatatgt atgctta		17
SEQ ID NO: 355 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 355 tgctcatttt ggcaaat		17
SEQ ID NO: 356 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 356 ttctcattcc tgcaagt		17
SEQ ID NO: 357 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 357 ttgctcattt tggcaaa		17
SEQ ID NO: 358 FEATURE misc_feature	moltype = DNA length = 17 Location/Qualifiers 1..17 note = Antisense Oligonucleotide	
source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 358 tttgcattt ttggcaa		17

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SEQ ID NO: 359	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 359		
atttgctcat ttggca		17
SEQ ID NO: 360	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 360		
cttgctcat ggcagaa		17
SEQ ID NO: 361	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 361		
ccttgctca tggcaga		17
SEQ ID NO: 362	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 362		
gccttgctc atggcag		17
SEQ ID NO: 363	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 363		
ggccttgct catggca		17
SEQ ID NO: 364	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 364		
tggccttgct tcatggc		17
SEQ ID NO: 365	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 365		
gtggccttg ctcatgg		17
SEQ ID NO: 366	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	

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SEQUENCE: 366	organism = synthetic construct	
agtggccttt gctcatg		17
SEQ ID NO: 367	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 367		
aagtggcctt tgctcat		17
SEQ ID NO: 368	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
source	note = Antisense Oligonucleotide	
	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 368		
aaagtggcct ttgctca		17
SEQ ID NO: 369	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 369		
cctttgctca tggcagaa		18
SEQ ID NO: 370	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 370		
ggcctttgct catggcag		18
SEQ ID NO: 371	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 371		
tggcctttgc tcatggca		18
SEQ ID NO: 372	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 372		
gtggcctttg ctcatggc		18
SEQ ID NO: 373	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
source	note = Antisense Oligonucleotide	
	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 373		
aaagtggcct ttgctcat		18
SEQ ID NO: 374	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	

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misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 374		
tggcctttgc tcatggcag		19
SEQ ID NO: 375	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = Antisense Oligonucleotide	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 375		
gtggcctttg ctcatggca		19
SEQ ID NO: 376	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 376		
gtggcctttg ctcatggcag		20
SEQ ID NO: 377	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 377		
ttttgttgca ggtcttc		17
SEQ ID NO: 378	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 378		
atattgttgc aggtcttc		17
SEQ ID NO: 379	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 379		
accatcttgt tgaaggt		17
SEQ ID NO: 380	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Antisense Oligonucleotide	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 380		
atattgttgc aggtcttc		18
SEQ ID NO: 381	moltype = DNA length = 17	
FEATURE	Location/Qualifiers	
misc_feature	1..17	
	note = Antisense Oligonucleotide	
source	1..17	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 381		

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gtctgctcgt ataccta	17
SEQ ID NO: 382	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 382	
taatattggt gggctgt	17
SEQ ID NO: 383	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 383	
ctaaaagtgg tggcctc	17
SEQ ID NO: 384	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 384	
ataatattgg tgggctg	17
SEQ ID NO: 385	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 385	
gctaaaagtg gtggcct	17
SEQ ID NO: 386	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 386	
aggggcagac tgtgaag	17
SEQ ID NO: 387	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 387	
taggggcaga ctgtgaa	17
SEQ ID NO: 388	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide
source	1..17
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 388	
agggctagac aaagctg	17
SEQ ID NO: 389	moltype = DNA length = 17
FEATURE	Location/Qualifiers
misc_feature	1..17
	note = Antisense Oligonucleotide

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source	1..17 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 389 gatagaaatg gtgaaag		17
SEQ ID NO: 390 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Antisense Oligonucleotide	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 390 taggggcaga ctgtgaag		18
SEQ ID NO: 391 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
misc_feature	1..20 note = Antisense Oligonucleotide	
SEQUENCE: 391 tcctgatgtt ccaccagatt		20
SEQ ID NO: 392 FEATURE source	moltype = DNA length = 19 Location/Qualifiers 1..19 mol_type = other DNA organism = synthetic construct	
misc_feature	1..19 note = Antisense Oligonucleotide	
SEQUENCE: 392 gcattattcc caccaggtc		19
SEQ ID NO: 393 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
misc_feature	1..20 note = Antisense Oligonucleotide	
SEQUENCE: 393 agactatcca tccaagtgct		20
SEQ ID NO: 394 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
misc_feature	1..20 note = Antisense Oligonucleotide	
SEQUENCE: 394 tggcccaaat catctacctc		20
SEQ ID NO: 395 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
misc_feature	1..20 note = Antisense Oligonucleotide	
SEQUENCE: 395 gcccactatt atcctagaca		20
SEQ ID NO: 396 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
misc_feature	1..20 note = Antisense Oligonucleotide	
SEQUENCE: 396 tgcttctgcc ctacaatttc		20

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SEQ ID NO: 397	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
SEQUENCE: 397		
tcccatatct ccagctctgct		20
SEQ ID NO: 398	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
SEQUENCE: 398		
gggtagattc actacatcca		20
SEQ ID NO: 399	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
SEQUENCE: 399		
gtctctaggt ttccactctc		20
SEQ ID NO: 400	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
misc_feature	1..20	
	note = Antisense Oligonucleotide	
SEQUENCE: 400		
gatcatcacc cttaccagct		20

What is claimed is:

1. A composition for treating pain, the composition comprising:

an oligonucleotide that hybridizes to locations on two RNAs that encode two different sodium channel proteins along a segment of each RNA that is at least about 75% complementary to one of SEQ ID NOs: 166-400 to thereby prevent translation of the RNA its respective sodium channel protein.

2. The composition of claim 1, wherein the oligonucleotide hybridizes to, and knocks down expression of, NaV1.7 and NaV1.8 pre-mRNA or mRNA.

3. The composition of claim 1, wherein a sequence of bases in the oligonucleotide has at least 89% identity to one of SEQ ID NOs: 166-400.

4. The composition of claim 1, wherein a sequence of bases in the oligonucleotide is at least 94% identical to one of SEQ ID NOs: 166-400, wherein the oligonucleotide can hybridize to, and induce RNase cleavage of, NaV1.7 and NaV1.8 pre-mRNA or mRNA.

5. The composition of claim 1, wherein the composition comprises a plurality of therapeutic oligonucleotides each having a base sequence at least 94% identical to one of SEQ ID NOs: 166-400, wherein each of the therapeutic oligonucleotides has a gapmer structure that comprises a central DNA segment flanked by modified RNA wings.

6. The composition of claim 1, wherein the oligonucleotide comprises two wings flanking a central region of at least 9 DNA bases.

7. The composition of claim 1, wherein at least one end of the oligonucleotide comprises modified RNA bases.

8. The composition of claim 7, wherein each modified RNA base is 2'-O-methoxyethyl RNA.

9. The composition of claim 1, wherein the oligonucleotide comprises about 19 bases.

10. The composition of claim 1, wherein the oligonucleotide has a backbone comprising a plurality of phosphorothioate bonds.

11. The composition of claim 1, wherein the oligonucleotide has a backbone comprising a mixture of phosphorothioate (PS) and phosphodiester (PO), in which the outermost inter-base linkages and all linkages involving a DNA base are PS, and in which the other three inter-RNA linkages of each wing comprise either two or three PS linkages, balance PO

12. The composition of claim 1, wherein the oligonucleotide has a base sequence that has been screened and determined to not meet a threshold match for any non-target transcripts in humans.

13. The composition of claim 1, wherein the oligonucleotide has a base sequence of one of SEQ ID NOs: 142-152 or 153-164.

14. The composition of claim 1, wherein when the composition is delivered to the dorsal root ganglion (DRG) neurons in vitro, the DRG neurons exhibit a dose-dependent knockdown of NaV1.7 and NaV1.8.

15. The composition of claim 1, wherein the oligonucleotide has a base sequence with at least a 94% match to one

of SEQ ID NO: 166-400, in which at least the outer-most inter-base linkages and all linkages involving a DNA base are phosphorothioate, and in which the oligonucleotide further comprises a central 9 DNA bases flanked by a 5' RNA wing and a 3' RNA wing, the 5' wing and the 3' wing each comprising five consecutive 2' modified RNA bases.

16. The composition of claim **51**, in which the oligonucleotide has the sequence of one of SEQ ID NO: 166-400 and in which both RNA wings consist of 5 2-methoxyethyl-modified RNA bases.

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